
SOFTWARE REQUIREMENT SPECIFICATION

for

LKFORUM WEBSITE

Version 1.0

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1 Introduction

1.1 Purpose

This SRS describes the functional and nonfunctional requirements for software release 1.0 of the Community Social Network Website (LKForum). This document is intended to be used by the members of the project team who will implement and verify the correct functioning of the system. Unless otherwise noted, all requirements specified here are committed for release 1.0

1.2 Intended Audience and Reading Suggestions

This SRS is for developers, project managers, users and testers. Further the discussion will provide all the internal, external, functional and also non-functional informations about "LKFORUM WEBSITE".

1.3 Project Scope

This SRS describes the functional and nonfunctional requirements for software release 1.0 of the Community Social Network Website (LKForum). This document is intended to be used by the members of the project team who will implement and verify the correct functioning of the system. Unless otherwise noted, all requirements specified here are committed for release 1.0

1.4 References

2 Overall Description

2.1 Product Perspective

Community Social Network LKForum is a new website that provides a comprehensive, modern solution for building and managing online communities. The context diagram in Figure 1 illustrates the external entities and system interfaces for release 1.0. The system is expected to evolve over several releases, ultimately connecting to third-party social media APIs for authentication (Google, Facebook) and expanding to include advanced content monetization tools and mobile-native applications.

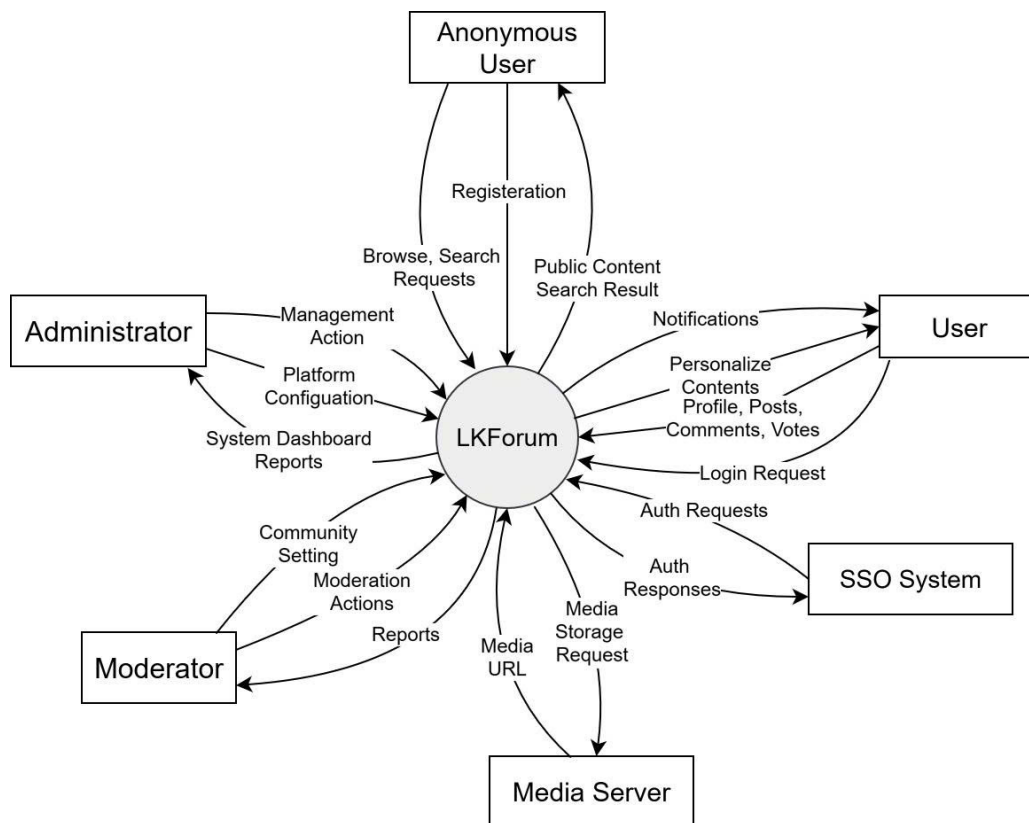


Figure 2.1: Context diagram for release 1.0 of LKForum

2.2 Operating Environment

- OE-1:** LKForum must operate correctly on modern web browsers including: Google Chrome (latest version), Mozilla Firefox (latest version), Apple Safari (latest version), and Microsoft Edge (latest version).
- OE-2:** Users can access LKForum via the Internet from desktop computers, laptops, and tablets. The interface must have responsive design to be compatible with various screen sizes.

2.3 Design and Implementation Constraints

- CO-1:** The user interface (front-end) source code must be developed using the Svelte framework.
- CO-2:** The system must use a standard MongoDB NoSQL database with a back-end developed using the Go framework.
- CO-3:** All HTML code must adhere to the HTML 5.0 standard.

2.4 Assumptions and Dependencies

- AS-1:** It is assumed that users have a stable Internet connection to access and use the platform's features.
- DE-1:** The social media login feature (Google, Facebook) depends on the APIs and policies of these third-party service providers. Any changes on their part may affect the operation of this feature.

3 System Features

The LKForum website is being built using the Go programming language for the back-end, Svelte with Vite for the front-end, MongoDB for primary data storage, and Redis for caching and session management.

Back-End – Go with the Gin web framework.

Front-End – Svelte with Vite.

Database – MongoDB.

Cache and Session Store – Redis.

3.1 Use Case Description

3.1.1 UC-01 - Sign Up

Name	Sign Up
Description	This use case describes how a new user creates an account and verifies the email address using OTP.
Actor	User
Trigger	When the user clicks the “Sign Up” button.
Pre-condition	The user is not logged in to the system. The user is on the sign up page.
Post-condition	The user account is created and verified. The user is authenticated and can access the system.

Table 3.1: Use Case Specification - Sign Up

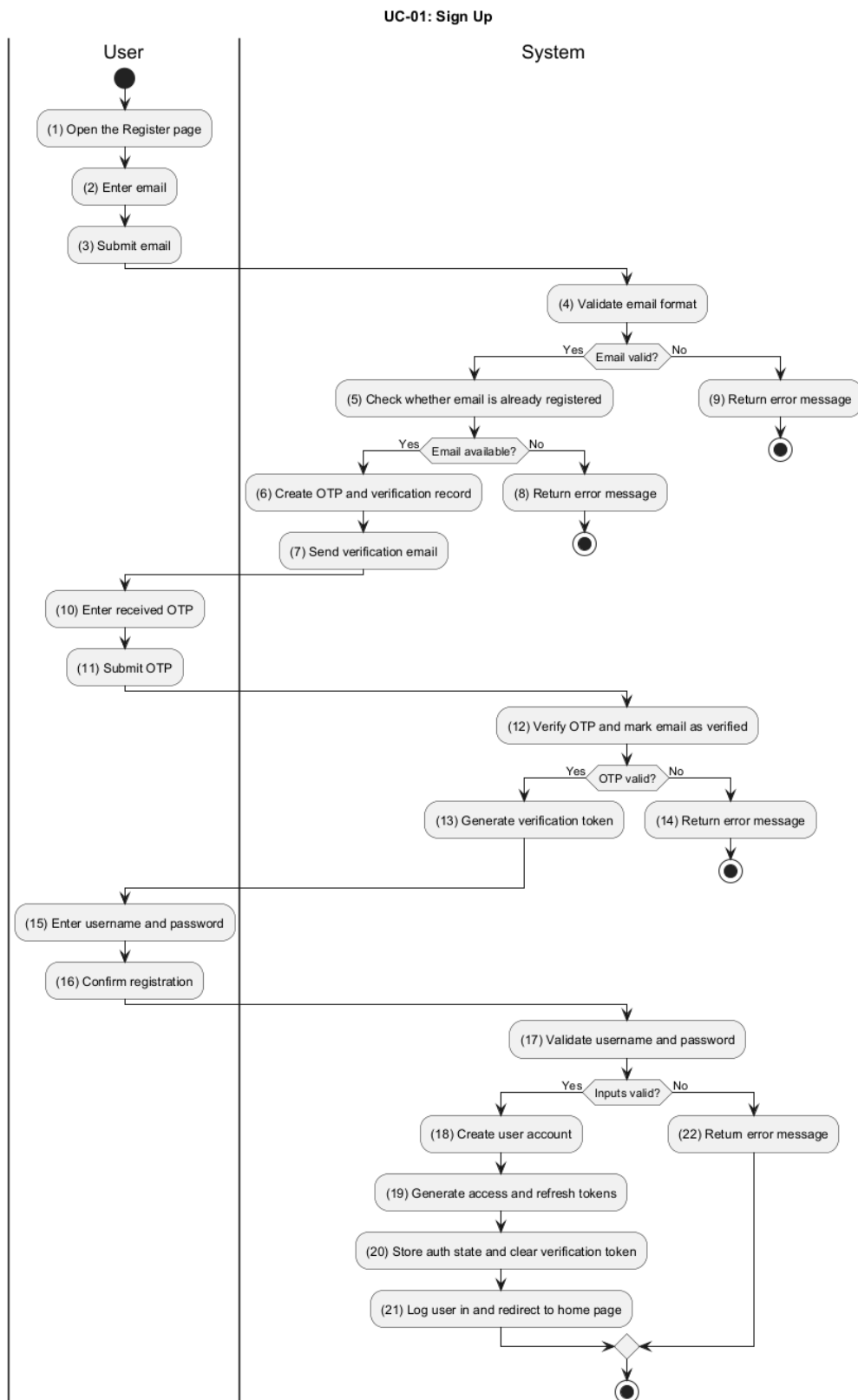


Figure 3.1: Activity Diagram for UC-01 Sign Up

ID	Activity	Rule Type	Description
BR-UC01-01	(4)	Validation Rules	Validate Email Format: - The system checks that email is present and valid. - If the email is missing or invalid, the system returns a bad-request error.
BR-UC01-02	(5)	Constraint Rules	Check Email Availability and Prepare Verification Record: - The system checks whether the email already exists in the user repository. - If the email already exists, the system returns an email-exists error.
BR-UC01-03	(7)	Message Rules	Send Verification Email: - Successful verification-code delivery shall use MSG-1 or MSG-9; resending shall use MSG-2 or MSG-10. - OTP validation failures shall return ERR-17 or ERR-18. - Completed verification and registration shall use MSG-4 or MSG-12; invalid registration data shall return ERR-30, ERR-21, ERR-22, or ERR-23.
BR-UC01-04	(12)	Validation Rules	Verify OTP and Issue Token: - The system checks that email and otp are present. - If no verification record exists, the system returns an invalid OTP error. - If the verification record is already marked verified, the system returns an email-already-verified error. - If the OTP does not match or has expired, the system returns an invalid or expired OTP error. - Otherwise, the system marks the verification record as verified and generates a verification token from the verified email and nonce.

Table 3.2: Business Rules for UC-01 - Sign Up

ID	Activity	Rule Type	Description
BR-UC01-05	(17)	Validation Rules	Validate Registration Fields: - The system checks that username is between 3 and 20 characters. - The system checks that password is at least 8 characters long and strong enough.
BR-UC01-06	(18-20)	Workflow Rules	Create Account and Complete Login: - The system hashes the password with <i>GenerateFromPassword()</i> before storing it. - The system checks that the username is still available and that the email is still not registered. - The system creates the new local user with default settings, deletes the verification record, invalidates the username cache, and generates access and refresh tokens using <i>GenerateToken()</i>.
BR-UC01-07	(21)	Redirect Rules	Redirect After Registration: The system logs the user in and redirects to the home page.

Table 3.2: Business Rules for UC-01 - Sign Up

3.1.2 UC-02 - Sign In

Name	Sign In
Description	This use case describes how a registered user logs into the system using local credentials or Google OAuth.
Actor	User
Trigger	When the user clicks the “Sign In” button.
Pre-condition	The user is not logged in to the system. The user is on the sign in page.
Post-condition	The user is authenticated. The user is redirected to the appropriate page.

Table 3.3: Use Case Specification - Sign In

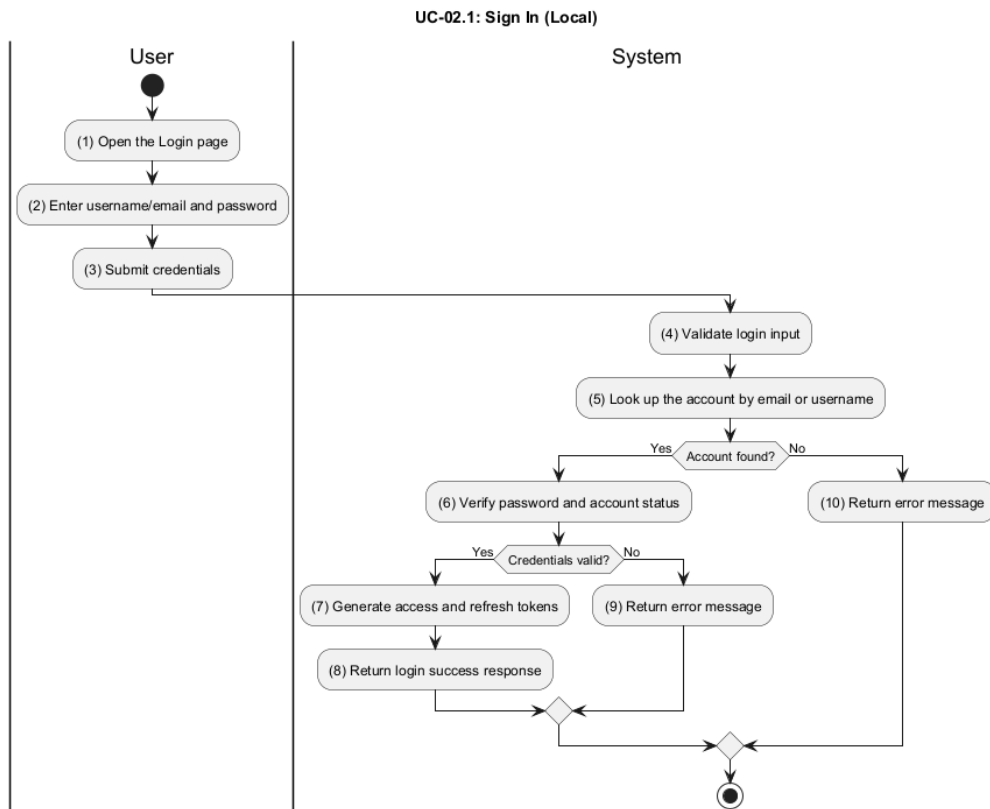


Figure 3.2: Activity Diagram for UC-02 Local Sign In

ID	Activity	Rule Type	Description
BR-UC02-01	(4)	Validation Rules	Validate Login Input: - The controller checks the JSON body for identifier and password in UserLoginRequest. - If binding fails or either field is missing, the controller returns a bad request response. - The controller does not validate the identifier format here; the service decides whether to search by email or username.

Table 3.4: Business Rules for UC-02.1 - Sign In (Local)

ID	Activity	Rule Type	Description
BR-UC02-02	(5)	Workflow Rules	Look Up Account: - The service checks whether identifier is an email using <i>isEmail()</i>. - The service looks up the account by email or username using the user repository. - If no account exists, the service returns an invalid credentials error.
BR-UC02-03	(6)	Authorization Rules	Verify Password and Account Status: - The system checks that the account uses the local provider. - The service compares the stored password with password using <i>bcrypt.CompareHashAndPassword()</i>. - If the password hash is empty or the comparison fails, the service returns an invalid credentials error. - If IsVerified is false, the service returns an email-not-verified error. - If the user is banned and the ban has not expired, the service returns a user-inactive error.
BR-UC02-04	(7–10)	Message Rules	Return Authentication Response: - Successful local sign-in shall use MSG-3. - Failed local sign-in shall return ERR-1. - Google sign-in and setup failures shall return ERR-2, ERR-19, or ERR-30 as applicable.

Table 3.4: Business Rules for UC-02.1 - Sign In (Local)

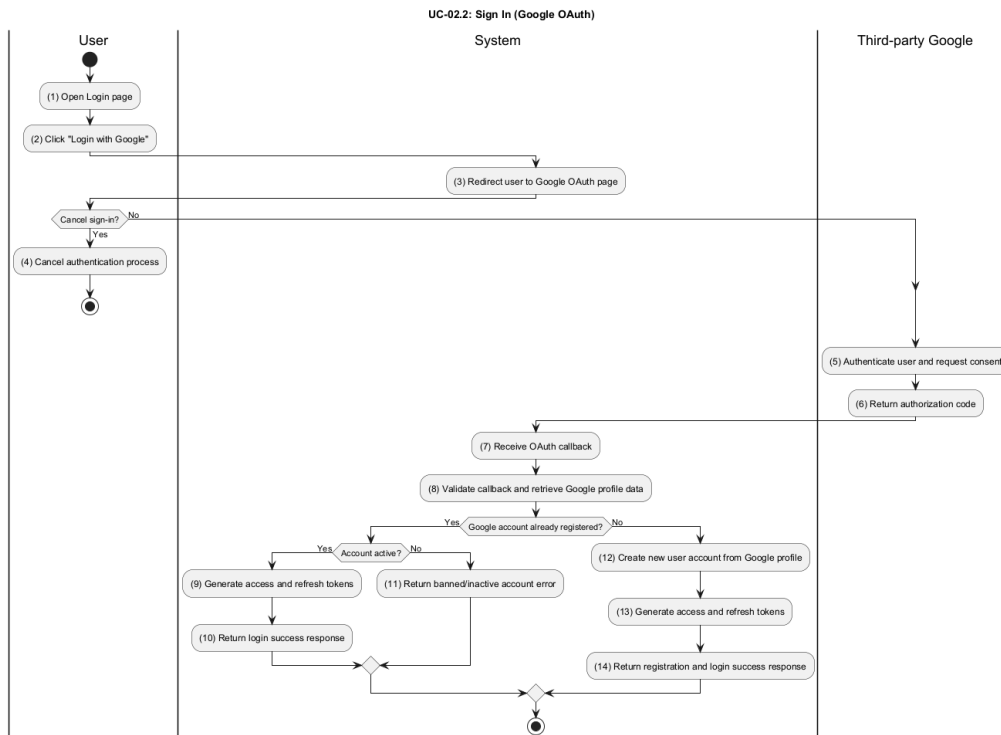


Figure 3.3: Activity Diagram for UC-02 Google Sign In

ID	Activity	Rule Type	Description
BR-UC02-05	(3)	Redirect Rules	Start Google OAuth Flow: - The system redirects the user to Google OAuth using a generated state value. - Google authenticates the user and returns an authorization code.

Table 3.5: Business Rules for UC-02.2 - Sign In (Google OAuth)

ID	Activity	Rule Type	Description
BR-UC02-06	(8)	Validation Rules	Process Google Callback: • The controller checks that the callback contains code. • If code is missing, the controller redirects to the frontend error page with a missing-auth-code message. • The service exchanges code for Google profile data using <i>GetGoogleUserInfo()</i>. • If Google rejects the authorization code or profile lookup fails, the controller redirects to the frontend error page with the returned error message.
BR-UC02-07	(9–11)	Authorization Rules	Login or Setup Decision: • If an account exists and provider is Google, the service generates access and refresh tokens using <i>GenerateToken()</i>. • If the account belongs to a different login method, the service returns a login-method-mismatch error. • If the account is banned and the ban has not expired, the service returns a user-inactive error. • If no Google account exists, the service creates a setup token using <i>CreateSetupToken()</i> and returns the setup-required result.

Table 3.5: Business Rules for UC-02.2 - Sign In (Google OAuth) (Continued)

ID	Activity	Rule Type	Description
BR-UC02-08	(12)	Validation Rules	Validate Google Setup Input: - The controller checks the JSON body for setup_token and username in CompleteGoogleSetupRequest. - If binding fails or setup_token is missing, the controller returns a bad request response. - The service validates that username is at least 3 characters and no more than 20 characters. - If username already exists, the service returns a username-exists error. - If the Google email is already taken, the service returns an email-exists error.
BR-UC02-09	(13)	Workflow Rules	Create Google Account: - The system parses setup_token to recover the Google claims. - The system creates the user with provider set to Google, stores provider_id, applies default settings, and copies the Google avatar URL when available. - The system generates access and refresh tokens after the user is created.
BR-UC02-10	(14)	Message Rules	Return Setup Success: - Successful Google-account completion shall use MSG-3 after the account and tokens are issued. - Validation failures during setup shall return ERR-30, ERR-21, ERR-22, or ERR-23 when relevant.

Table 3.5: Business Rules for UC-02.2 - Sign In (Google OAuth)

3.1.3 UC-03 - Forgot Password

Name	Forgot Password
Description	This use case describes how a user resets a forgotten password through email verification and OTP.
Actor	User
Trigger	When the user clicks the “Forgot Password” link.
Pre-condition	The user has an existing account and can access the registered email address. The user is not logged in to the system.
Post-condition	The user password is reset successfully. The user can sign in again with the new password.

Table 3.6: Use Case Specification - Forgot Password

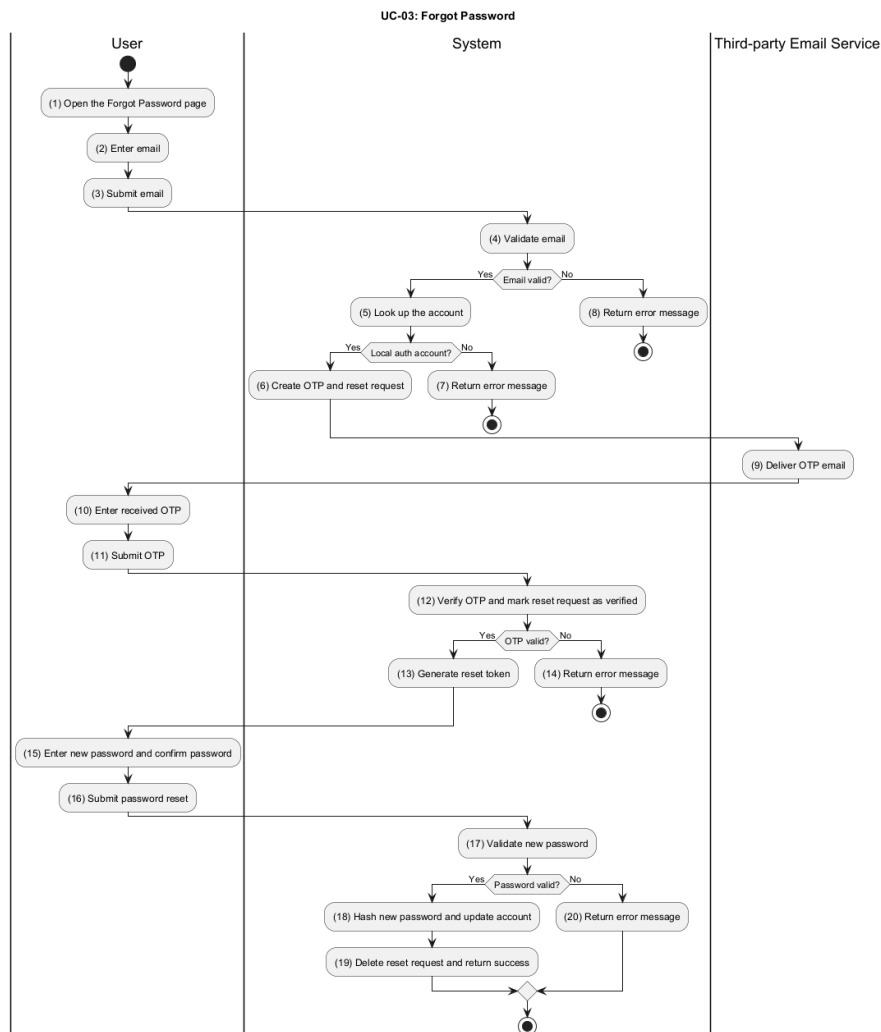


Figure 3.4: Activity Diagram for UC-03

ID	Activity	Rule Type	Description
BR-UC03-01	(4)	Validation Rules	Validate Reset Request: - The controller checks email in <code>ForgotPasswordRequest</code>. - If binding fails or email is missing or invalid, the controller returns a bad request response.
BR-UC03-02	(5)	Authorization Rules	Validate Account Ownership: - The service looks up the account by email using the user repository. - If the account does not exist, the service returns an email-not-registered error. - If the account exists but the provider is not local, the service returns a login-method-mismatch error.
BR-UC03-03	(6)	Workflow Rules	Create Password Reset Request: - The system deletes any existing reset record for the same email. - The system generates a 6-digit OTP, a nonce, and an expiry timestamp using <i><code>generateOTP()</code></i> and <i><code>generateNonce()</code></i>. - The system stores a new password-reset record with IsVerified = false.
BR-UC03-04	(7)	Message Rules	Return Reset Request Result: - Successful reset-code delivery shall use MSG-6. - If the account is not local, the service shall return ERR-19; unknown local accounts shall return ERR-20.

Table 3.7: Business Rules for UC-03 - Forgot Password

ID	Activity	Rule Type	Description
BR-UC03-05	(12)	Validation Rules	Validate Reset OTP: - The controller checks email and otp in <code>VerifyResetPasswordOTPRequest</code>. - If binding fails, the controller returns a bad request response. - If no reset record exists for the email, the service returns an invalid OTP error. - If the reset record is already verified, the service returns an email-already-verified error. - If the OTP does not match or has expired, the service returns an invalid or expired OTP error.
BR-UC03-06	(12)	Workflow Rules	Verify OTP and Generate Reset Token: - The service marks the reset record as verified. - The service generates a reset token from the verified email and nonce using <i>CreateVerificationToken()</i>.
BR-UC03-07	(14)	Message Rules	Return OTP Error: - Successful reset-OTP verification shall use MSG-7. - Invalid or expired reset OTP values shall return ERR-17 or ERR-18.

Table 3.7: Business Rules for UC-03 - Forgot Password (Continued)

ID	Activity	Rule Type	Description
BR-UC03-08	(17)	Validation Rules	Validate New Password: - The controller checks reset_token and new_password in ResetPasswordRequest. - If binding fails or reset_token is missing, the controller returns a bad request response. - The DTO binding requires new_password to be present and at least 6 characters long. - The service requires new_password to be at least 8 characters long. - The service requires new_password to contain uppercase, lowercase, numeric, and special characters.
BR-UC03-09	(18)	Workflow Rules	Hash and Update Password: - The service parses the reset token and verifies the stored nonce. - The service looks up the user by email. - The service hashes new_password with <i>GenerateFromPassword()</i>. - The service updates the user password in the repository.
BR-UC03-10	(19)	Workflow Rules	Clean Up Reset Record: - The service deletes the reset record after a successful password change. - The controller returns a success response telling the user they can log in with the new password.
BR-UC03-11	(20)	Message Rules	Return Password Error: - Successful password reset shall use MSG-8 or MSG-13. - Invalid replacement passwords shall return ERR-22 or ERR-23.

Table 3.7: Business Rules for UC-03 - Forgot Password

3.1.4 UC-04 - Manage Profile

Name	Manage Profile
Description	This use case describes how a user updates profile information such as bio, avatar, and cover image.
Actor	User
Trigger	When the user opens the profile edit page.
Pre-condition	The user is logged in to the system. The user has permission to edit the profile.
Post-condition	The profile information is updated and stored. The new profile data is visible across the system.

Table 3.8: Use Case Specification - Manage Profile

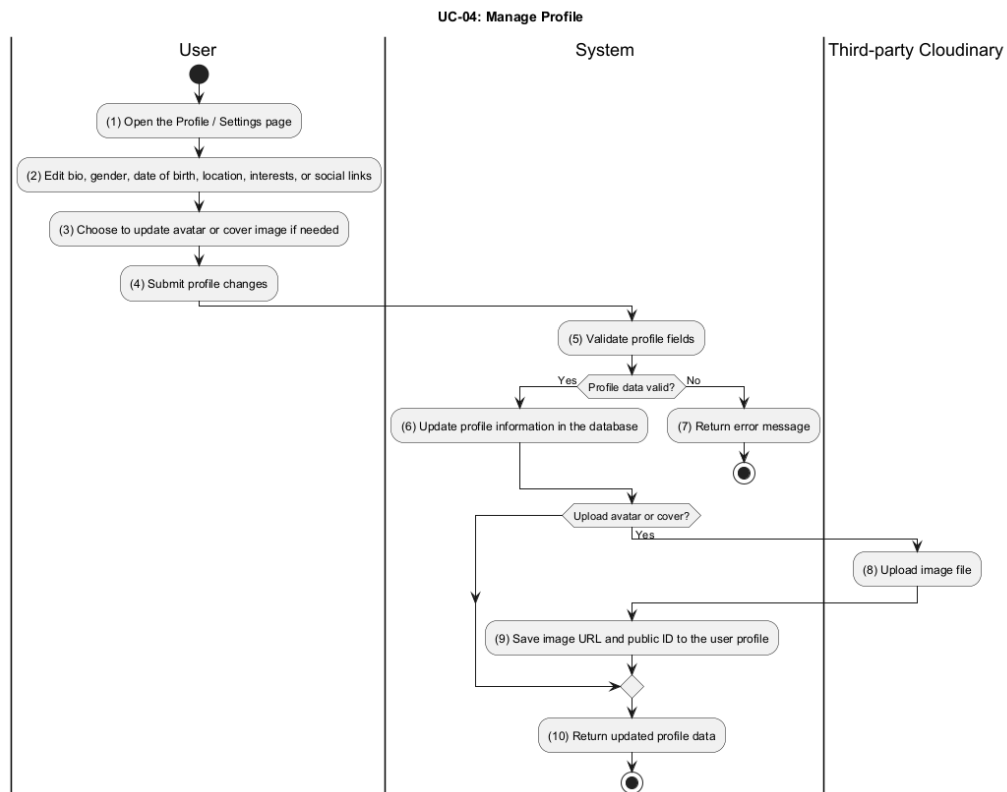


Figure 3.5: Activity Diagram for UC-04

Activity	BR Code	Description
(5)	BR1	Validate Profile Fields: - The system checks the submitted profile fields before saving. bio is limited by DTO validation, gender must match a valid model value, and date_of_birth must parse as a valid date. - The system validates location against allowed provinces and ensures interests do not exceed the maximum allowed count.
(6)	BR2	Update Profile Data: - The system loads the current user and updates RoleContent.AsUser. - Empty string values clear the corresponding profile field. - Invalid gender, location, or interest values return an error instead of being saved.
(7–8)	BR3	Upload and Save Profile Images: - When the user uploads an avatar or cover image, the system sends the file to <i>cloudinary.UploadImages()</i>. - The system stores the returned image URL and PublicID through <i>UpdateAvatar()</i> or <i>UpdateCover()</i>. - If an older image exists, the system deletes the old asset asynchronously using <i>cloudinary.Delete()</i>.
(9)	BR4	Return Updated Profile: The system returns the refreshed user profile after saving changes.
(6–9)	BR5	Message Rules: - Successful profile and avatar actions shall use MSG-48, MSG-53, MSG-55, and the confirmation prompt MSG-57. - Failed profile updates shall use MSG-49, MSG-54, or MSG-56. - Validation failures shall return ERR-28, ERR-32, ERR-33, ERR-35, ERR-36, ERR-37, or ERR-38.

Table 3.9: Business Rules for UC-04 - Manage Profile

3.1.5 UC-05 - Change Settings

Name	Change Settings
Description	This use case describes how a user changes account preferences such as privacy, notifications, and theme.
Actor	User
Trigger	When the user opens the settings page.
Pre-condition	The user is logged in to the system. The settings page is available to the user.
Post-condition	The selected preferences are saved. The account settings are updated for future sessions.

Table 3.10: Use Case Specification - Change Settings

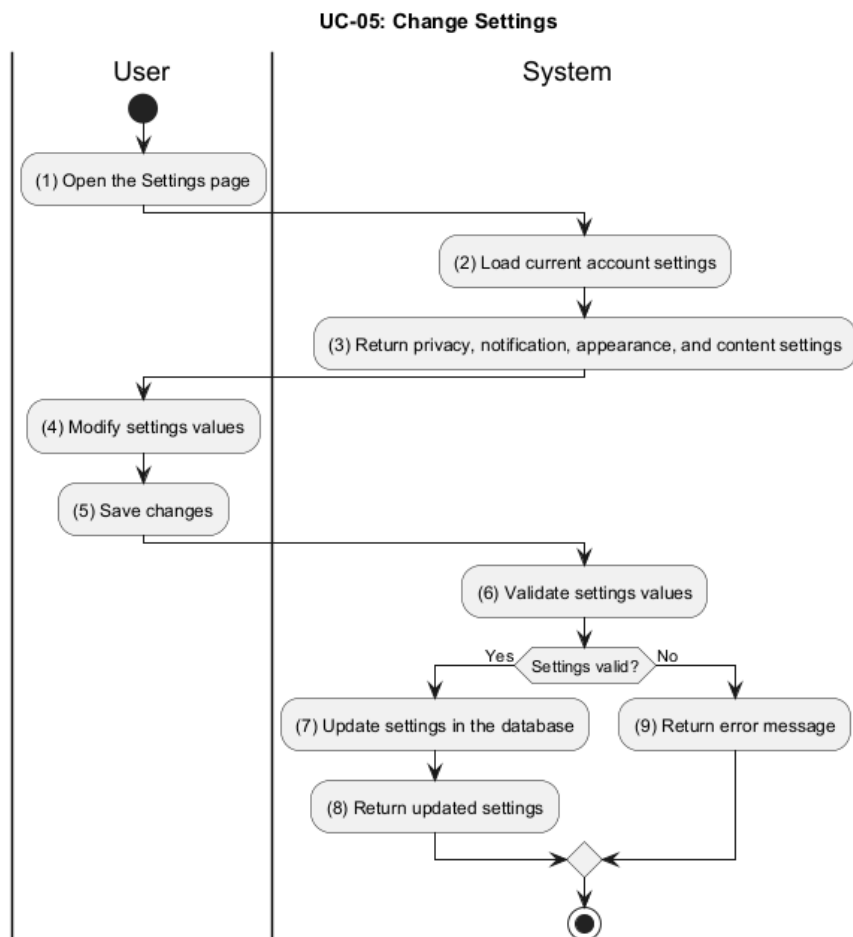


Figure 3.6: Activity Diagram for UC-05

Activity	BR Code	Description
(2-3)	BR1	Load Current Settings: - The system loads the user's current Settings from the database using <i>GetSettings()</i>. - If the user has no settings record, the system returns default settings through <i>dto.FromSettings()</i>.
(6)	BR2	Validate Settings Values: - The system validates theme and fontSize using <i>model.IsValidTheme()</i> and <i>model.IsValidFontSize()</i>. - The system accepts only values supported by the model enums and returns a bad-request error for invalid input.
(7-8)	BR3	Persist Updated Settings: - The system updates Appearance, Notifications, Privacy, and Content fields in user.Settings. - The system saves the updated user record with <i>userRepo.Update()</i>. - The system returns the updated settings after persistence.
(6-8)	BR4	Message Rules: - Successful account-setting changes shall use MSG-51 or MSG-13. - Failed setting updates shall use MSG-52 or MSG-50. - Invalid input or authentication failures shall return ERR-14, ERR-2, ERR-10, or ERR-25.

Table 3.11: Business Rules for UC-05 - Change Settings

3.1.6 UC-06 - View Activity History

Name	View Activity History
Description	This use case describes how a user reviews past posts, comments, and interactions.
Actor	User
Trigger	When the user opens the activity history page.
Pre-condition	The user is logged in to the system. The user has activity records in the system.
Post-condition	The activity history is displayed to the user. The user can review prior activity.

Table 3.12: Use Case Specification - View Activity History

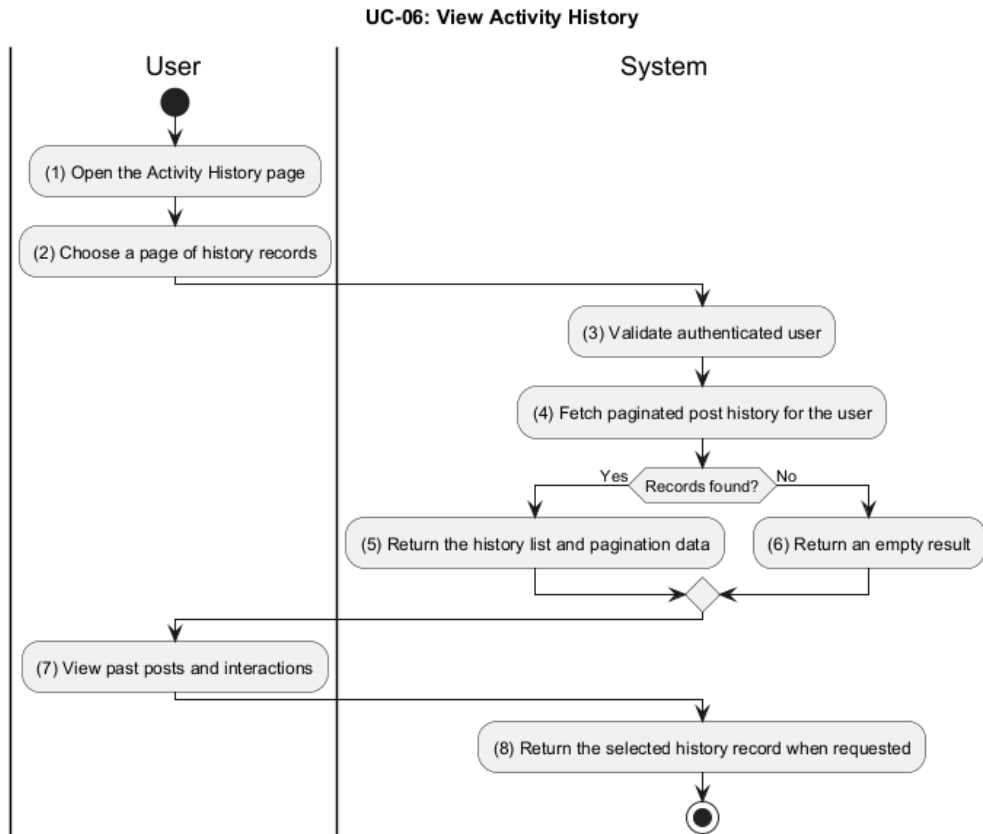


Figure 3.7: Activity Diagram for UC-06

Activity	BR Code	Description
(3-5)	BR1	Validate Ownership and Fetch History: - The system checks that the requester is authenticated and that userID matches the current user for the history request. - The system fetches paginated post history using <i>GetPostHistoryByUserID()</i>. - If no records are found, the system returns an empty result with pagination data.
(7)	BR2	Return Selected History Record: - The system returns a single history item only when the requester owns the record. - The system rejects access if requesterID does not match the stored UserID.

Table 3.13: Business Rules for UC-06 - View Activity History

Activity	BR Code	Description
(3-7)	BR3	Message Rules: • This use case has no dedicated success toast; successful requests return inline history data and empty-state results. • Access or paging failures shall return ERR-10, ERR-13, ERR-28, ERR-26, or ERR-27.

Table 3.13: Business Rules for UC-06 - View Activity History

3.1.7 UC-07 - Follow / Unfollow User

Name	Follow / Unfollow User
Description	This use case describes how a user subscribes to or unsubscribes from another user's activity feed.
Actor	User
Trigger	When the user clicks the follow or unfollow action on another profile.
Pre-condition	The user is logged in to the system. The target profile is available for follow actions.
Post-condition	The follow relationship is created or removed. The user's feed and follower lists are updated.

Table 3.14: Use Case Specification - Follow / Unfollow User

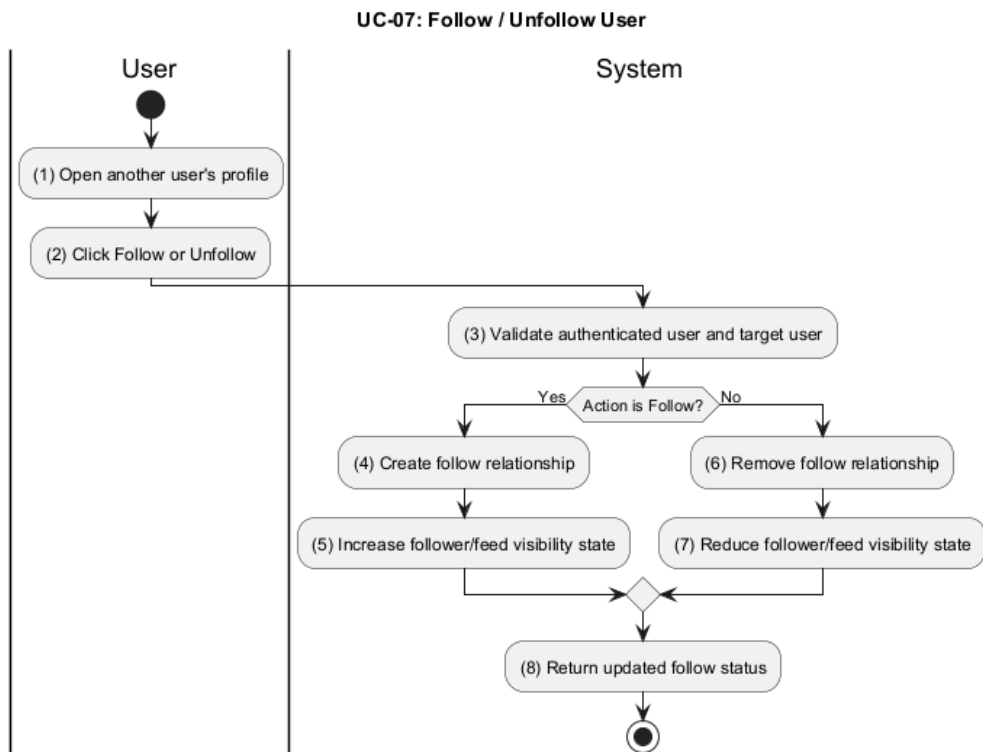


Figure 3.8: Activity Diagram for UC-07

Activity	BR Code	Description
(3–6)	BR1	Backend Rule Status: • No follow or unfollow route, controller, or service is present in the backend code. • The repository currently has no server-side business rules to define for this use case. • This use case remains documented-only until a backend implementation is added.
(3–6)	BR2	Message Rules: • No reusable application MSG or ERR contract is currently implemented for follow or unfollow actions in the codebase. • When this use case is implemented, it should define dedicated success messages and authorization or user-not-found errors consistent with the appendix scheme.

Table 3.15: Business Rules for UC-07 - Follow / Unfollow User

3.1.8 UC-08 - Browse Content

Name	Browse Content
Description	This use case describes how a visitor or user explores communities and public posts.
Actor	Guest User / User
Trigger	When the user opens the forum homepage or a community page.
Pre-condition	The browsing page is accessible. The user may be logged in or not logged in.
Post-condition	Public posts and community content are shown. The user can continue browsing or open a post.

Table 3.16: Use Case Specification - Browse Content

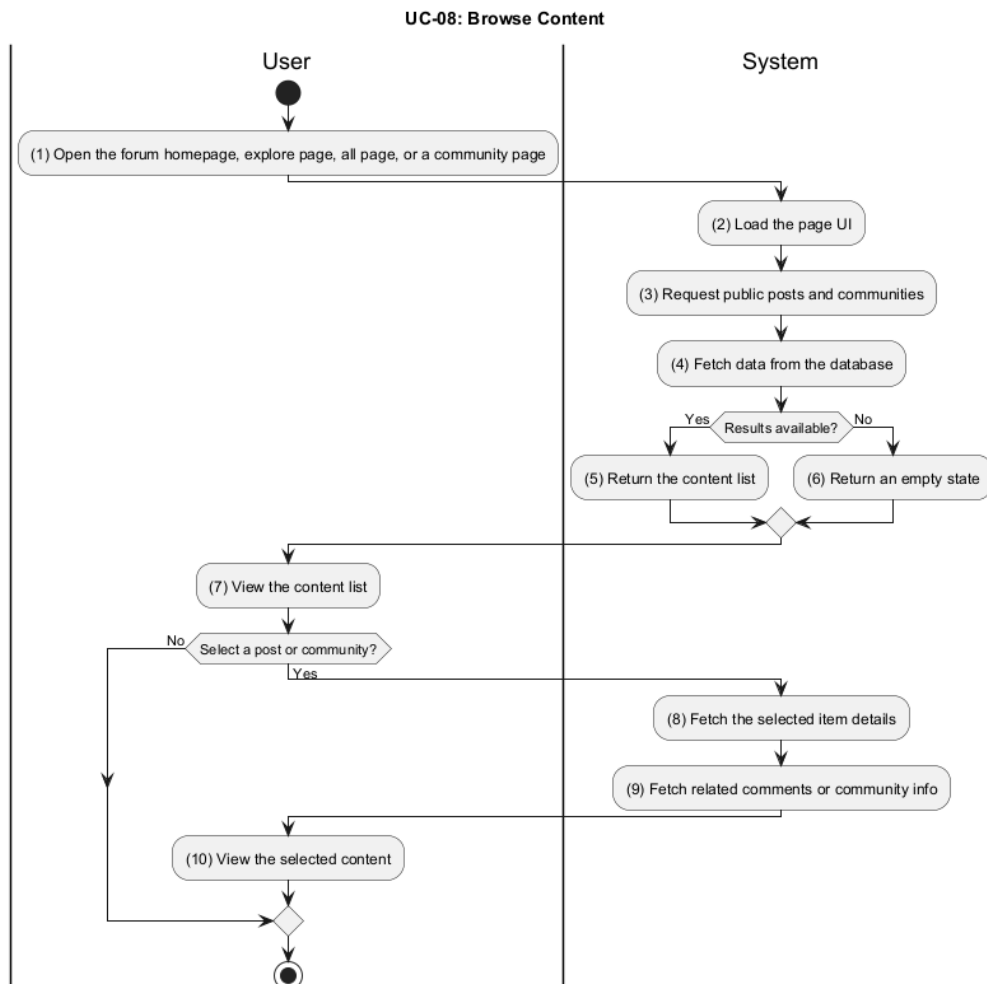


Figure 3.9: Activity Diagram for UC-08

Activity	BR Code	Description
(2–5)	BR1	Load Public Content: • The system loads the page UI and requests public posts and communities. • The system fetches content from the database using <i>GetPosts()</i> and community lookup functions. • If no results exist, the system returns an empty state instead of an error.
(7–9)	BR2	Open Selected Content: • If the user selects a post, the system loads the post details using <i>GetPostByID()</i>. • If the user selects a community, the system loads the community details using <i>GetCommunityByID()</i> or <i>GetCommunityByName()</i>. • The system also returns related comments or community information for the selected item.
(2–9)	BR3	Message Rules: • This use case has no dedicated success toast; successful browsing returns content lists or empty states directly in the UI. • Content-loading failures shall use MSG-58, MSG-84, or MSG-83. • Lookup failures shall return ERR-53, ERR-39, ERR-26, or ERR-27.

Table 3.17: Business Rules for UC-08 - Browse Content

3.1.9 UC-09 - Search Content

Name	Search Content
Description	This use case describes how a user searches for posts, users, or communities by keyword.
Actor	User
Trigger	When the user enters a search term and submits the search.
Pre-condition	The search page is available. The system has searchable content.
Post-condition	Matching results are returned to the user. The user can open a result or refine the search.

Table 3.18: Use Case Specification - Search Content

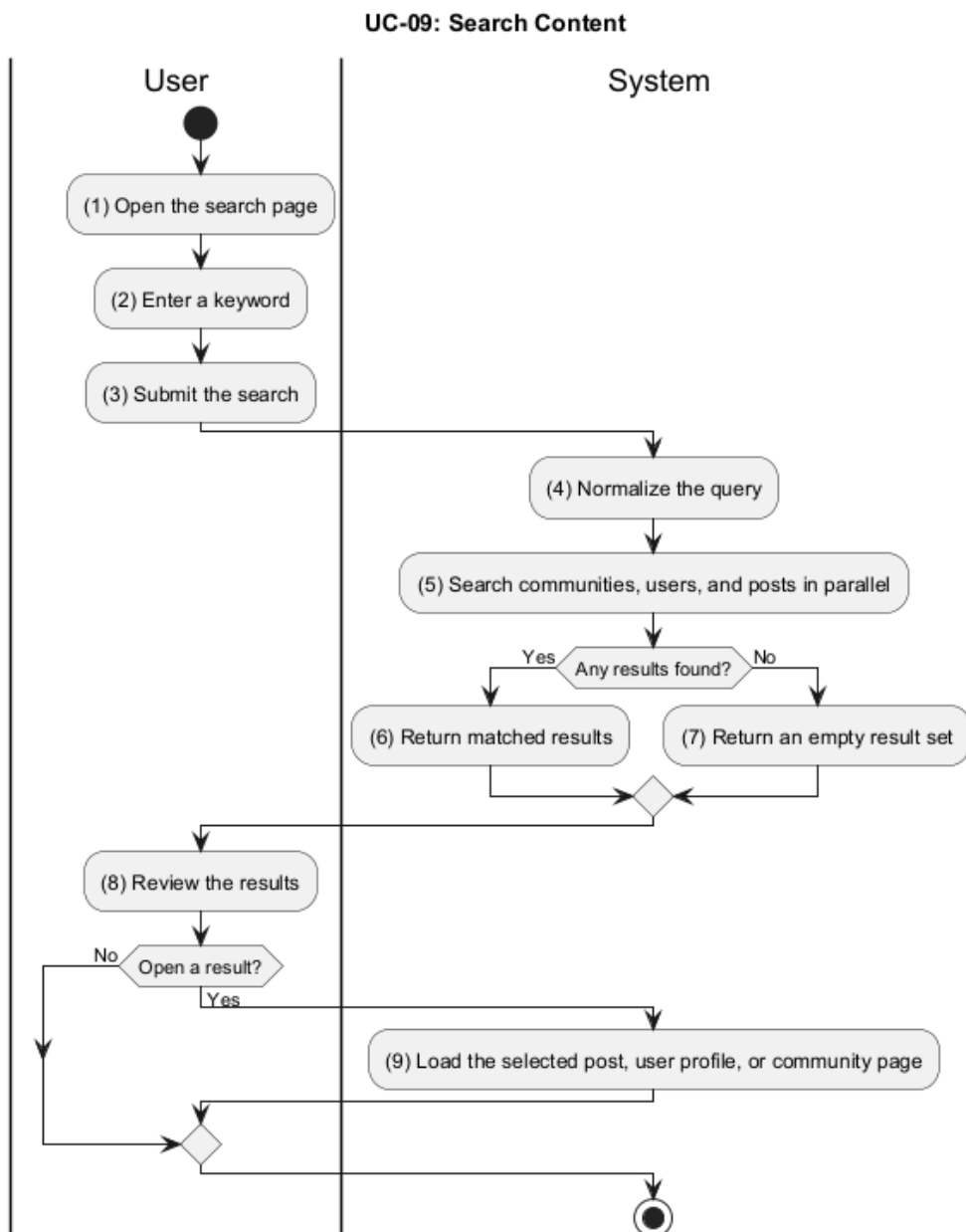


Figure 3.10: Activity Diagram for UC-09

Activity	BR Code	Description
(4–6)	BR1	Normalize and Search in Parallel: • The system normalizes the query before searching. • The system searches posts, users, and communities in parallel using <i>GetPosts()</i>, <i>GetUsers()</i>, and <i>GetCommunitiesFilter()</i>. • If no results are found, the system returns an empty result set.
(8)	BR2	Open Search Result: • If the user opens a result, the system loads the selected post, user profile, or community page. • The system uses the corresponding detail lookup function based on the result type.
(4–8)	BR3	Message Rules: • This use case has no dedicated success toast; successful searches return matching results or an empty result set inline. • Opening a result or resolving a target shall return ERR-53, ERR-28, ERR-39, ERR-26, or ERR-27 when the selected item cannot be resolved.

Table 3.19: Business Rules for UC-09 - Search Content

3.1.10 UC-10 - Create Post

Name	Create Post
Description	This use case describes how a member creates and submits a new post in a community.
Actor	User
Trigger	When the user clicks the “Create Post” action.
Pre-condition	The user is logged in to the system. The user is a member of the selected community.
Post-condition	The post is stored successfully. The post appears in the community or enters moderation if required.

Table 3.20: Use Case Specification - Create Post

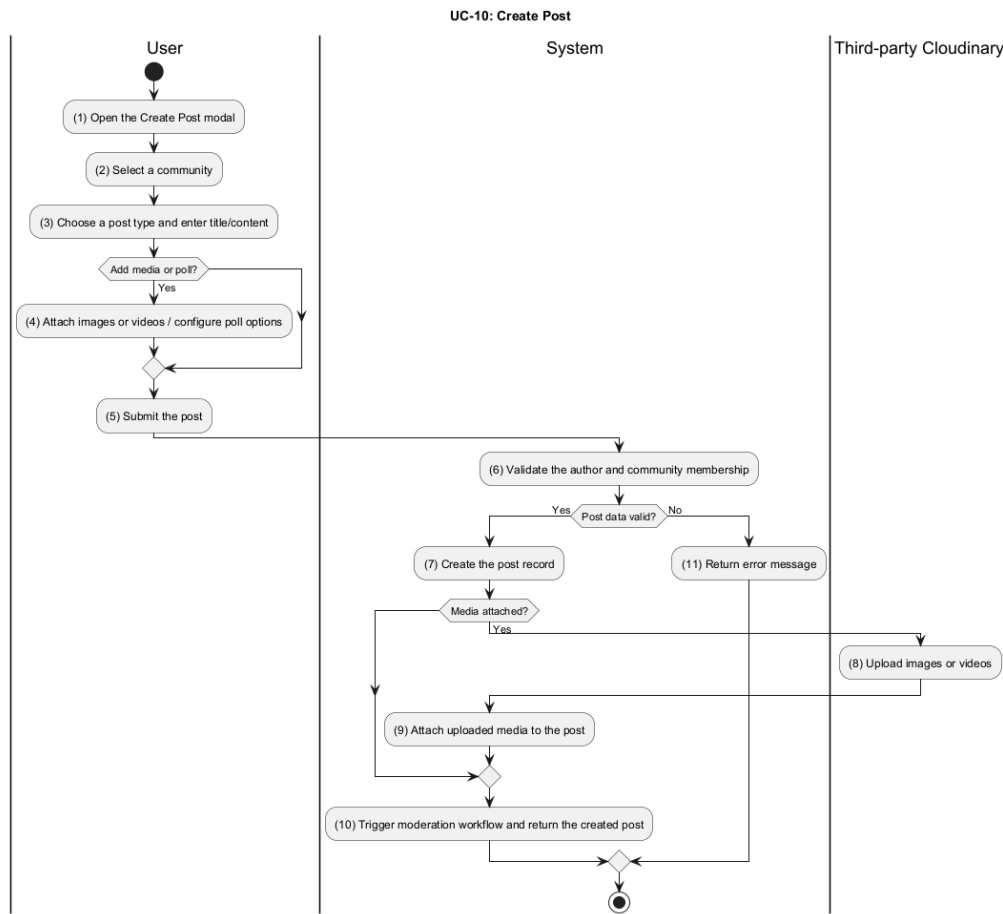


Figure 3.11: Activity Diagram for UC-10

Activity	BR Code	Description
(5)	BR1	Validate Author and Community: - The system checks that the user is authenticated and is a member of the selected community. - The system rejects the request if the community is banned or the user is banned from the community.
(6)	BR2	Create the Post Record: - The system creates the post with the submitted title, content, tags, and type. - The system sets the initial moderation state based on community approval rules and author role. - The system auto-upvotes the post for the creator using <i>VoteOnTargetByAuthor()</i>.

Table 3.21: Business Rules for UC-10 - Create Post

Activity	BR Code	Description
(7–8)	BR3	Handle Media and Poll Data: • If images or videos are attached, the system uploads them using <i>UploadImages()</i> or <i>UploadVideos()</i>. • If a poll is included, the system stores poll options and poll settings in the post content.
(9)	BR4	Return Created Post: • The system returns the created post after persistence and moderation setup are complete.
(5–9)	BR5	Message Rules: • Successful post creation returns the created post inline; when moderation or membership state blocks creation, the system should not fabricate a success message. • Validation and authorization failures shall return ERR-39, ERR-45, ERR-46, ERR-48, ERR-26, or ERR-10 as applicable.

Table 3.21: Business Rules for UC-10 - Create Post

3.1.11 UC-11 - Edit Post

Name	Edit Post
Description	This use case describes how a member modifies an existing post and preserves edit history.
Actor	User
Trigger	When the user opens an existing post and chooses “Edit”.
Pre-condition	The user is logged in to the system. The user owns the post or has edit permission.
Post-condition	The edited post is saved successfully. The edit history is retained.

Table 3.22: Use Case Specification - Edit Post

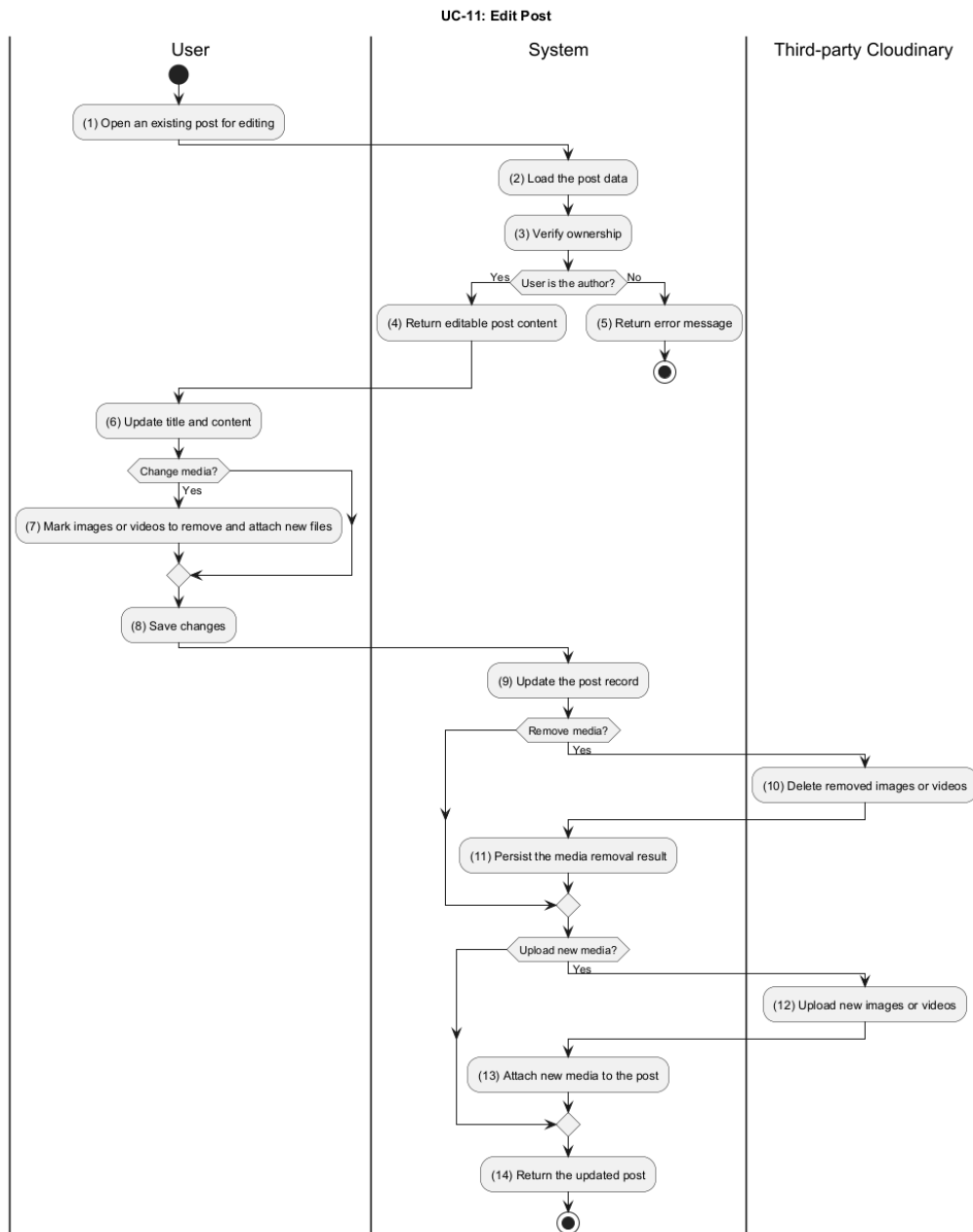


Figure 3.12: Activity Diagram for UC-11

Activity	BR Code	Description
(2-4)	BR1	Authorize Editing: - The system loads the post and verifies that the requester is the author. - If the requester is not the author, the system returns a forbidden error.

Table 3.23: Business Rules for UC-11 - Edit Post

Activity	BR Code	Description
(5–6)	BR2	Apply Content Changes: • The system updates title, text, and tags when they are provided. • If no fields change, the system returns the current post without modifying data.
(7–11)	BR3	Handle Media Updates: • If media is removed, the system deletes the old assets using <i>Delete()</i>. • If new media is uploaded, the system stores it using <i>UploadImages()</i> or <i>UploadVideos()</i>. • The system persists the updated media links back to the post record.
(12)	BR4	Return Updated Post: • The system returns the refreshed post after the update is saved.
(5–12)	BR5	Message Rules: • Successful post updates shall use MSG-29; cancellation of unsaved edits shall use MSG-20. • Failed updates shall use MSG-30. • Authorization or content-state failures shall return ERR-53, ERR-13, ERR-56, or ERR-26.

Table 3.23: Business Rules for UC-11 - Edit Post

3.1.12 UC-12 - Delete Post

Name	Delete Post
Description	This use case describes how a member or moderator removes a post from public view using soft delete.
Actor	User / Moderator
Trigger	When the user clicks the “Delete” action on a post.
Pre-condition	The user is logged in to the system. The user has permission to delete the post.
Post-condition	The post is marked as deleted. The post is hidden from public view.

Table 3.24: Use Case Specification - Delete Post

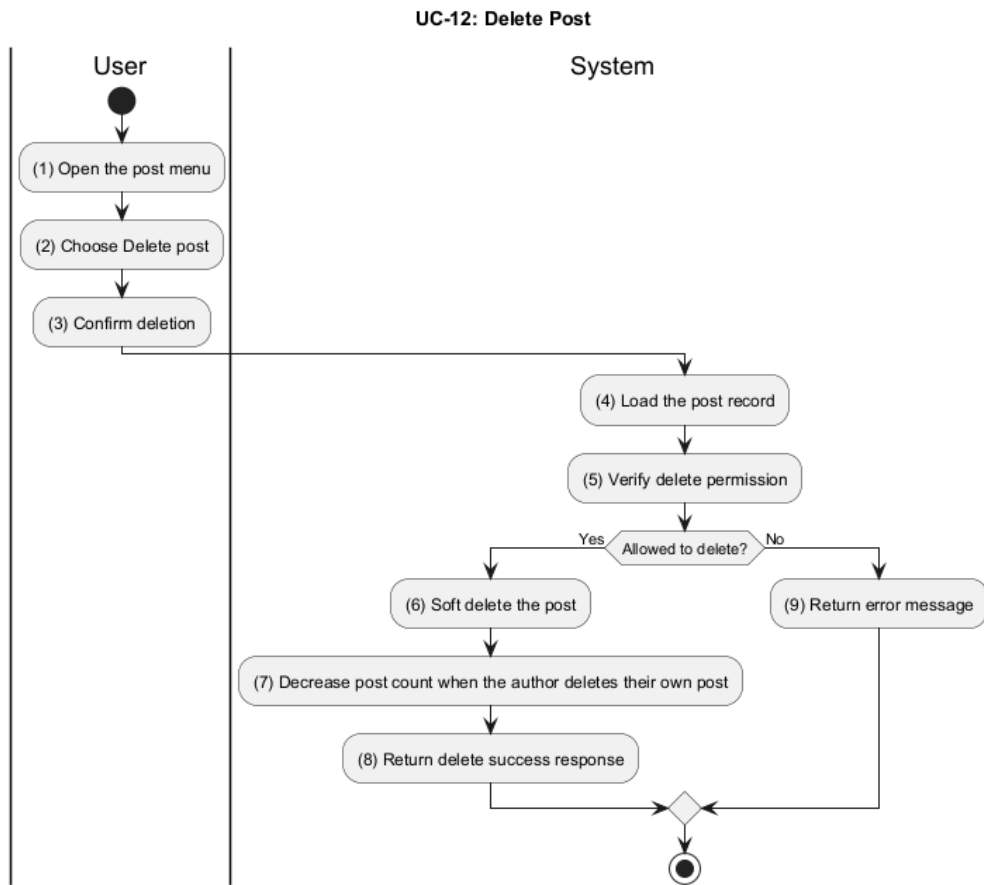


Figure 3.13: Activity Diagram for UC-12

Activity	BR Code	Description
(4-5)	BR1	Verify Delete Permission: - The system loads the post record and checks whether the requester is the author. - If the requester is not the author, the system checks for admin, creator, or moderator permission. - If no permission exists, the system returns a forbidden error.
(6-8)	BR2	Soft Delete the Post: - The system marks the post as deleted in storage. - If the author deletes their own post, the system decreases the author's post count. - The system returns a delete success response.

Table 3.25: Business Rules for UC-12 - Delete Post

Activity	BR Code	Description
(4-8)	BR3	Message Rules: • Delete confirmation shall use MSG-14; successful deletion shall use MSG-87. • Failed deletion shall use MSG-27. • Permission or lookup failures shall return ERR-53, ERR-13, or ERR-26.

Table 3.25: Business Rules for UC-12 - Delete Post

3.1.13 UC-13 - Post Comment

Name	Post Comment
Description	This use case describes how a user adds a direct comment to a post.
Actor	User
Trigger	When the user submits a comment on a post.
Pre-condition	The user is logged in to the system. The target post is visible and commentable.
Post-condition	The comment is created and attached to the post. The comment count is updated.

Table 3.26: Use Case Specification - Post Comment

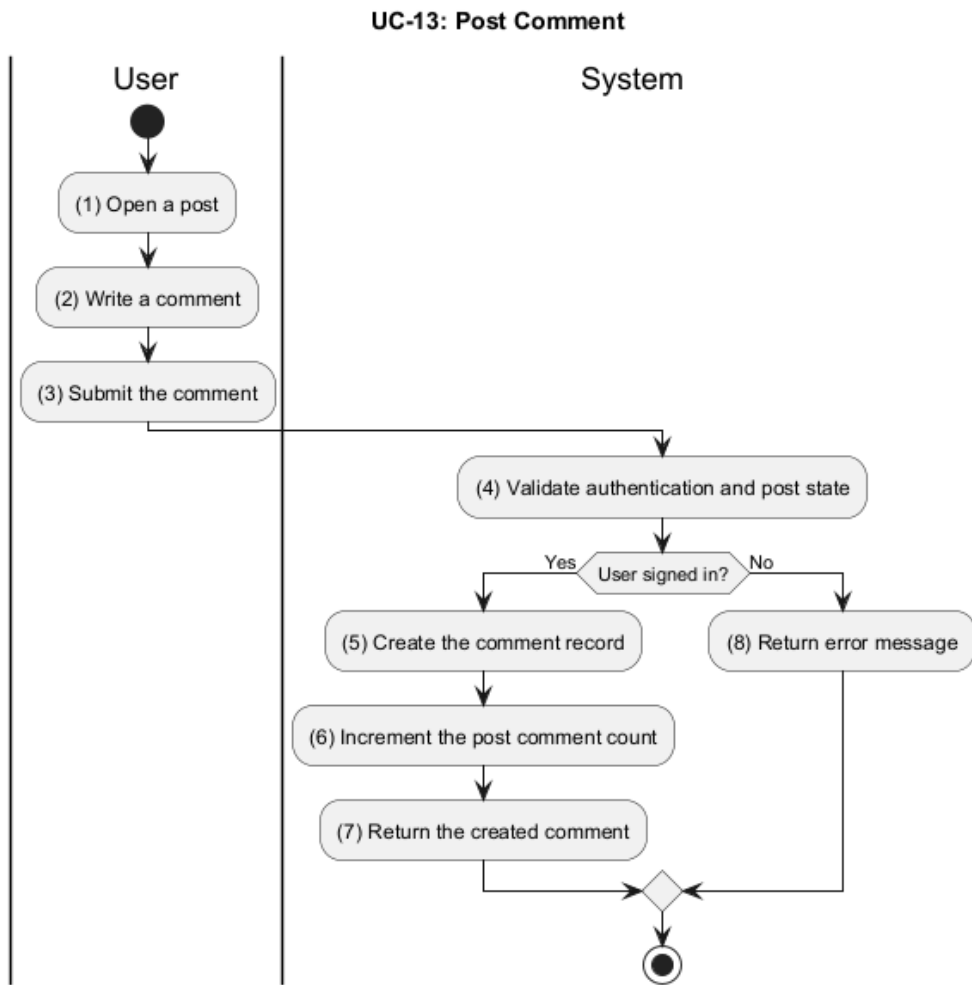


Figure 3.14: Activity Diagram for UC-13

Activity	BR Code	Description
(4)	BR1	Validate Post and Permissions: - The system validates authentication and checks that the target post exists. - The system blocks the comment if the user is banned or muted in the post's community.

Table 3.27: Business Rules for UC-13 - Post Comment

Activity	BR Code	Description
(5-7)	BR2	Create the Comment: • The system stores the new comment with the current user as author. • The system increments the post comment count and the user's comment count. • The system publishes the comment-created event for follow-up processing.
(4-7)	BR3	Message Rules: • Failed comment submission shall use MSG-64. • Validation and permission failures shall return ERR-53, ERR-61, ERR-45, ERR-10, or ERR-26.

Table 3.27: Business Rules for UC-13 - Post Comment

3.1.14 UC-14 - Reply to Comment

Name	Reply to Comment
Description	This use case describes how a user replies to an existing comment in a nested discussion thread.
Actor	User
Trigger	When the user submits a reply to a comment.
Pre-condition	The user is logged in to the system. The parent comment exists and can receive replies.
Post-condition	The reply is created as a nested comment. The discussion thread is updated.

Table 3.28: Use Case Specification - Reply to Comment

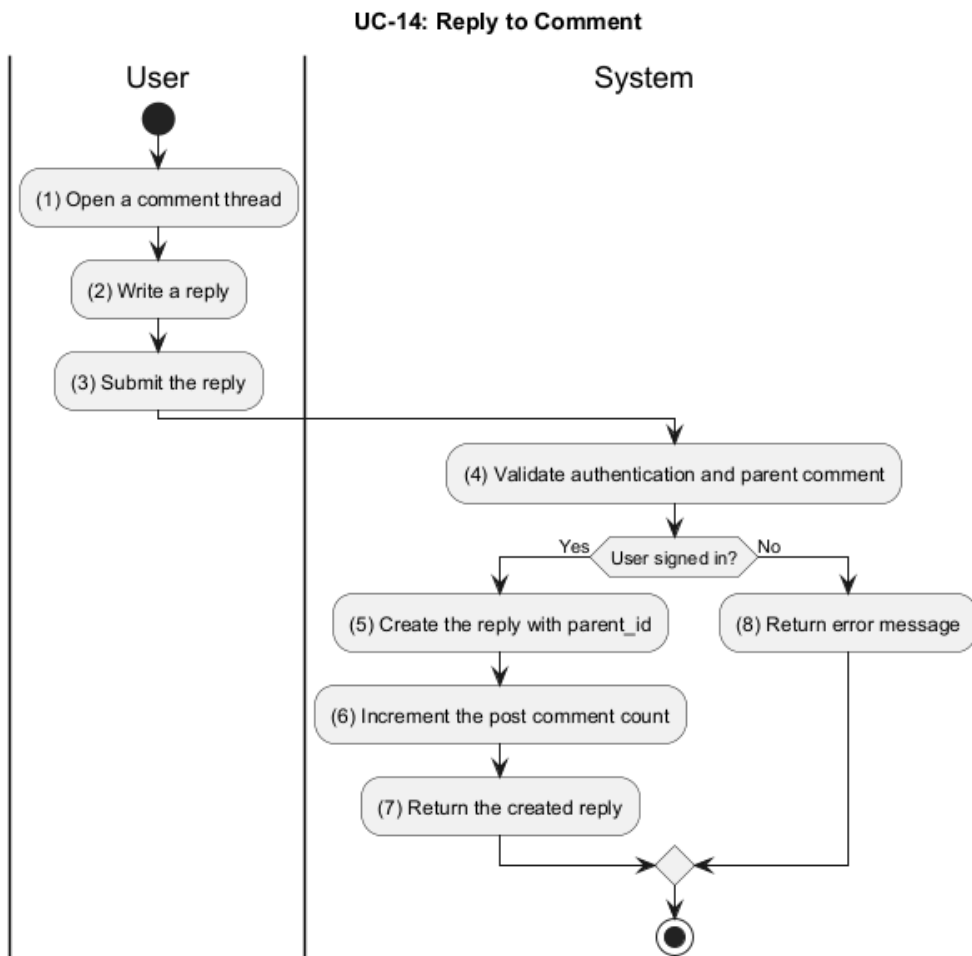


Figure 3.15: Activity Diagram for UC-14

Activity	BR Code	Description
(4)	BR1	Validate Parent Comment: - The system validates authentication and checks the parent comment when a reply is submitted. - The system loads the parent comment to derive the parent author for notification logic.
(5–7)	BR2	Create the Nested Reply: - The system stores the reply with a populated parent_id. - The system increments the post comment count and returns the created reply.

Table 3.29: Business Rules for UC-14 - Reply to Comment

Activity	BR Code	Description
(4-7)	BR3	Message Rules: - Failed reply submission shall use MSG-65. - Validation and permission failures shall return ERR-59, ERR-53, ERR-61, ERR-60, or ERR-10.

Table 3.29: Business Rules for UC-14 - Reply to Comment

3.1.15 UC-15 - Vote on Content

Name	Vote on Content
Description	This use case describes how a user upvotes or downvotes a post or comment.
Actor	User
Trigger	When the user selects an upvote or downvote action.
Pre-condition	The user is logged in to the system. The target content is available for voting.
Post-condition	The vote is recorded or updated. The content score is recalculated.

Table 3.30: Use Case Specification - Vote on Content

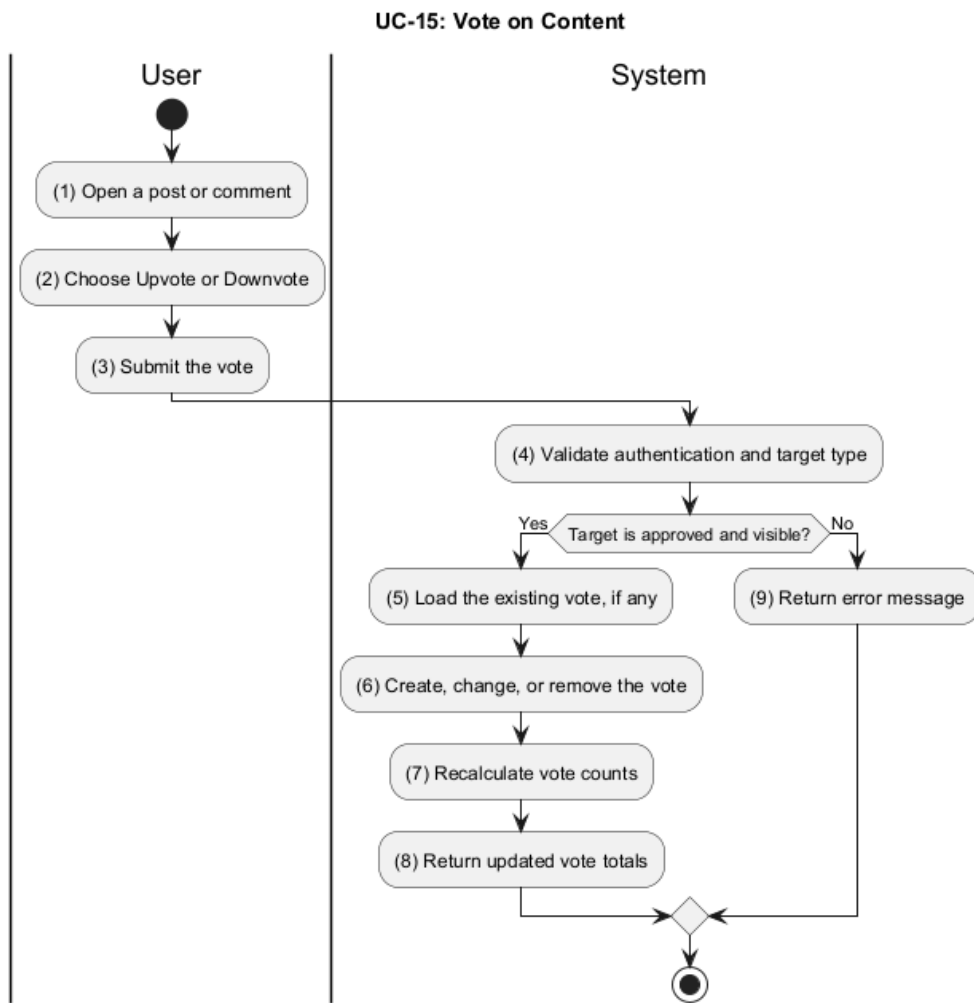


Figure 3.16: Activity Diagram for UC-15

Activity	BR Code	Description
(4)	BR1	Validate Voting Target: - The system validates authentication and checks the target type. - The system allows voting only when the post is approved or skipped, or when the target is a visible comment.

Table 3.31: Business Rules for UC-15 - Vote on Content

Activity	BR Code	Description
(5–8)	BR2	Apply Vote and Return Counts: • The system loads the existing vote using <i>GetUserVote()</i>. • The system creates, changes, or removes the vote using <i>VoteOnTarget()</i>. • The system recalculates vote counts and returns the updated totals.
(4–8)	BR3	Message Rules: • This use case updates vote state inline and does not define a dedicated success toast in the current UI. • Voting failures shall return ERR-53, ERR-59, ERR-54, ERR-13, or ERR-10.

Table 3.31: Business Rules for UC-15 - Vote on Content

3.1.16 UC-16 - Participate in Poll

Name	Participate in Poll
Description	This use case describes how a user casts a vote in a community poll.
Actor	User
Trigger	When the user submits a poll vote.
Pre-condition	The user is logged in to the system. The poll is open for voting.
Post-condition	The selected poll option is recorded. The poll results are updated.

Table 3.32: Use Case Specification - Participate in Poll

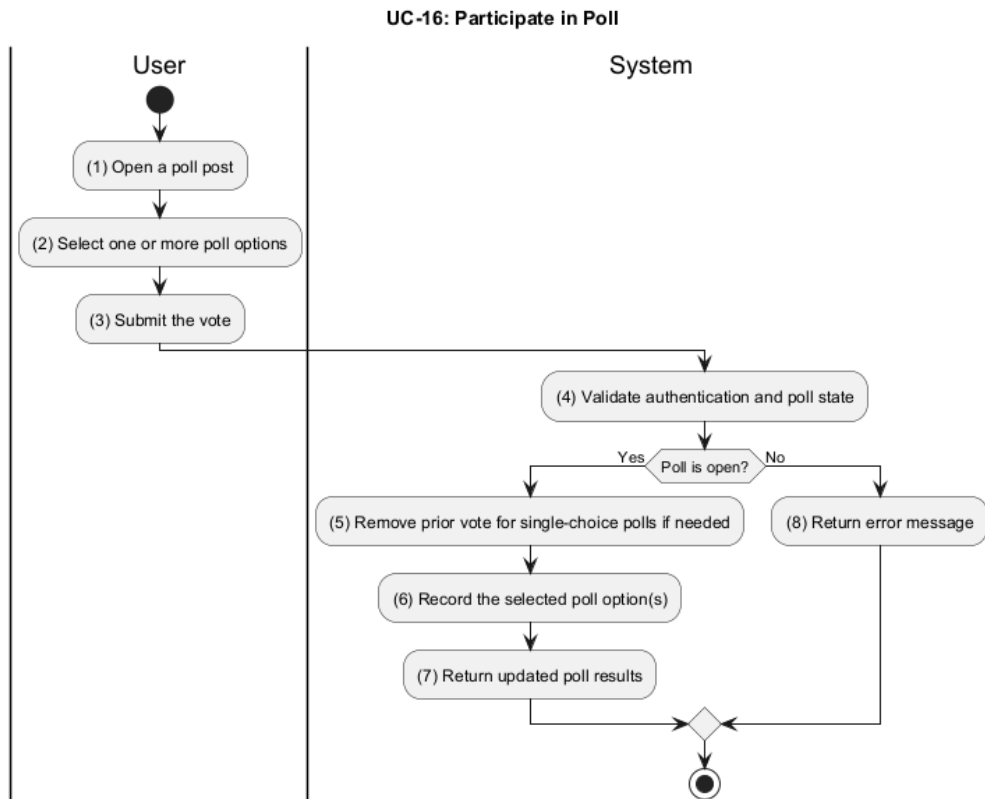


Figure 3.17: Activity Diagram for UC-16

Activity	BR Code	Description
(4)	BR1	Validate Poll State: - The system validates authentication and checks whether the poll is open. - The system rejects votes when the poll is closed or unavailable.
(5–7)	BR2	Record the Poll Vote: - The system removes a previous vote when the poll is single-choice. - The system records the selected poll option(s) using <i>VoteOnPoll()</i>. - The system returns the updated poll results to the client.

Table 3.33: Business Rules for UC-16 - Participate in Poll

Activity	BR Code	Description
(4-7)	BR3	Message Rules: • Poll-vote failures shall use MSG-25 or MSG-26. • Poll-state failures shall return ERR-53, ERR-54, ERR-55, ERR-56, or ERR-10.

Table 3.33: Business Rules for UC-16 - Participate in Poll

3.1.17 UC-17 - Save Post

Name	Save Post
Description	This use case describes how a user bookmarks a post for later viewing.
Actor	User
Trigger	When the user clicks the “Save” action on a post.
Pre-condition	The user is logged in to the system. The target post exists and is visible to the user.
Post-condition	The post is added to the user’s saved list. The saved state is available in the account.

Table 3.34: Use Case Specification - Save Post

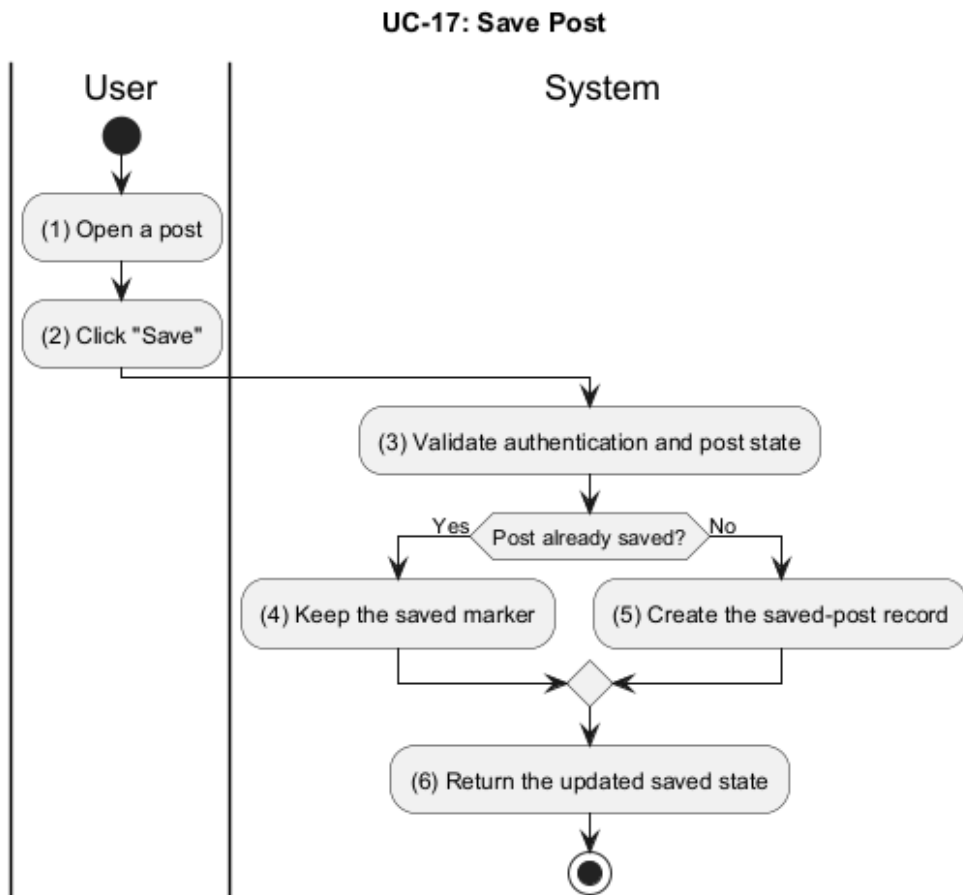


Figure 3.18: Activity Diagram for UC-17

Activity	BR Code	Description
(3-5)	BR1	Save or Unsave the Post: • The system validates authentication and checks that the post exists. • The system stores the saved-post relation using <i>SavePost()</i>. • If the post is already saved, the system keeps the saved marker instead of duplicating it.

Table 3.35: Business Rules for UC-17 - Save Post

Activity	BR Code	Description
(3-5)	BR2	Message Rules: • This use case updates the saved state inline in the UI and does not define a dedicated success toast in the current code. • Save or unsave failures shall return ERR-53, ERR-26, or ERR-10.

Table 3.35: Business Rules for UC-17 - Save Post

3.1.18 UC-18 - Hide Post

Name	Hide Post
Description	This use case describes how a user hides a post from their personal feed.
Actor	User
Trigger	When the user clicks the “Hide” action on a post.
Pre-condition	The user is logged in to the system. The target post is visible in the user’s feed.
Post-condition	The post is hidden from the user’s feed. The hidden-post record is stored for the account.

Table 3.36: Use Case Specification - Hide Post

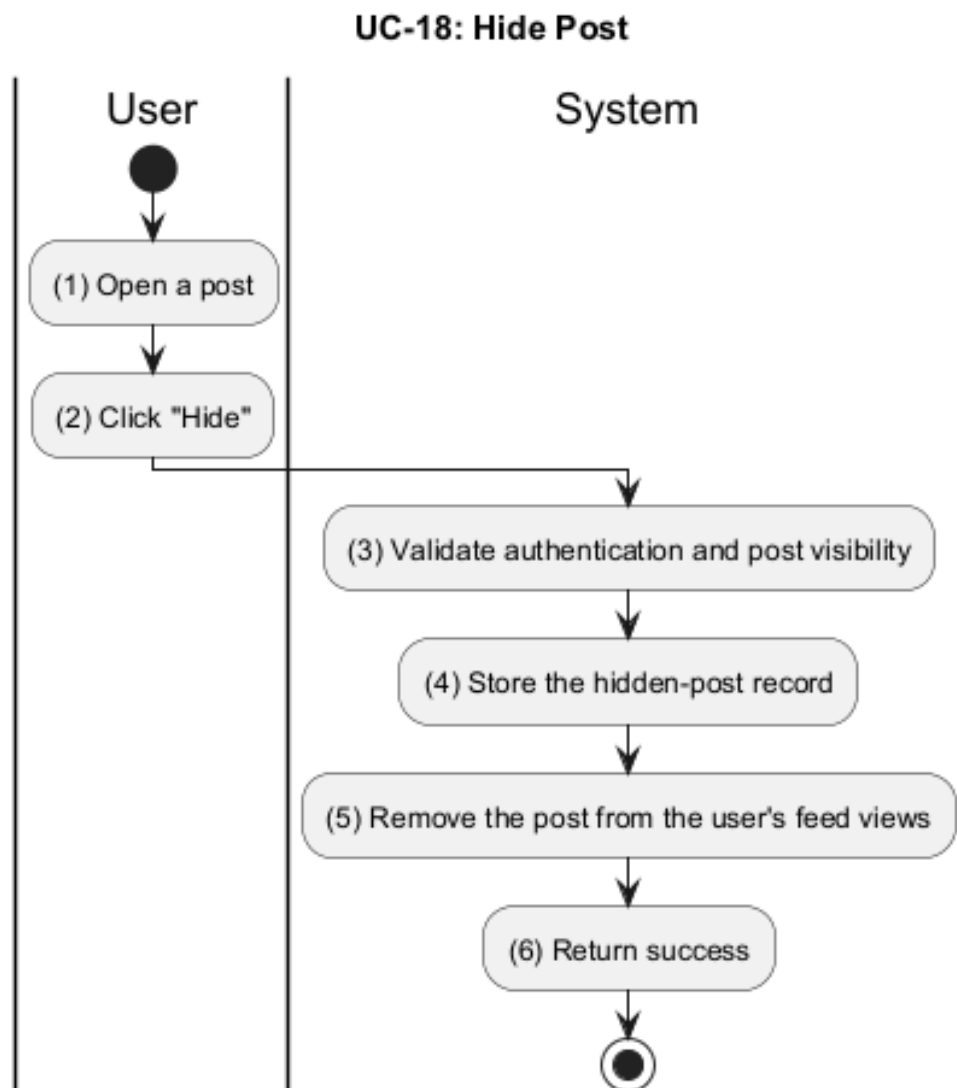


Figure 3.19: Activity Diagram for UC-18

Activity	BR Code	Description
(3–6)	BR1	Hide the Post: • The system validates authentication and confirms that the requester is the post author. • The system marks the post as hidden using <i>HidePost()</i>. • The system removes the post from the user’s visible feed views.

Table 3.37: Business Rules for UC-18 - Hide Post

Activity	BR Code	Description
(3-6)	BR2	Message Rules: • This use case updates hidden-post state inline and does not define a reusable success toast in the current UI. • Hide failures shall return ERR-53, ERR-13, ERR-26, or ERR-10.

Table 3.37: Business Rules for UC-18 - Hide Post

3.1.19 UC-19 - Manage Drafts

Name	Manage Drafts
Description	This use case describes how a user saves unfinished posts as drafts and publishes them later.
Actor	User
Trigger	When the user saves, edits, opens, or publishes a draft.
Pre-condition	The user is logged in to the system. The user has draft access.
Post-condition	The draft is saved, updated, or deleted after publishing. A published draft becomes a normal post.

Table 3.38: Use Case Specification - Manage Drafts

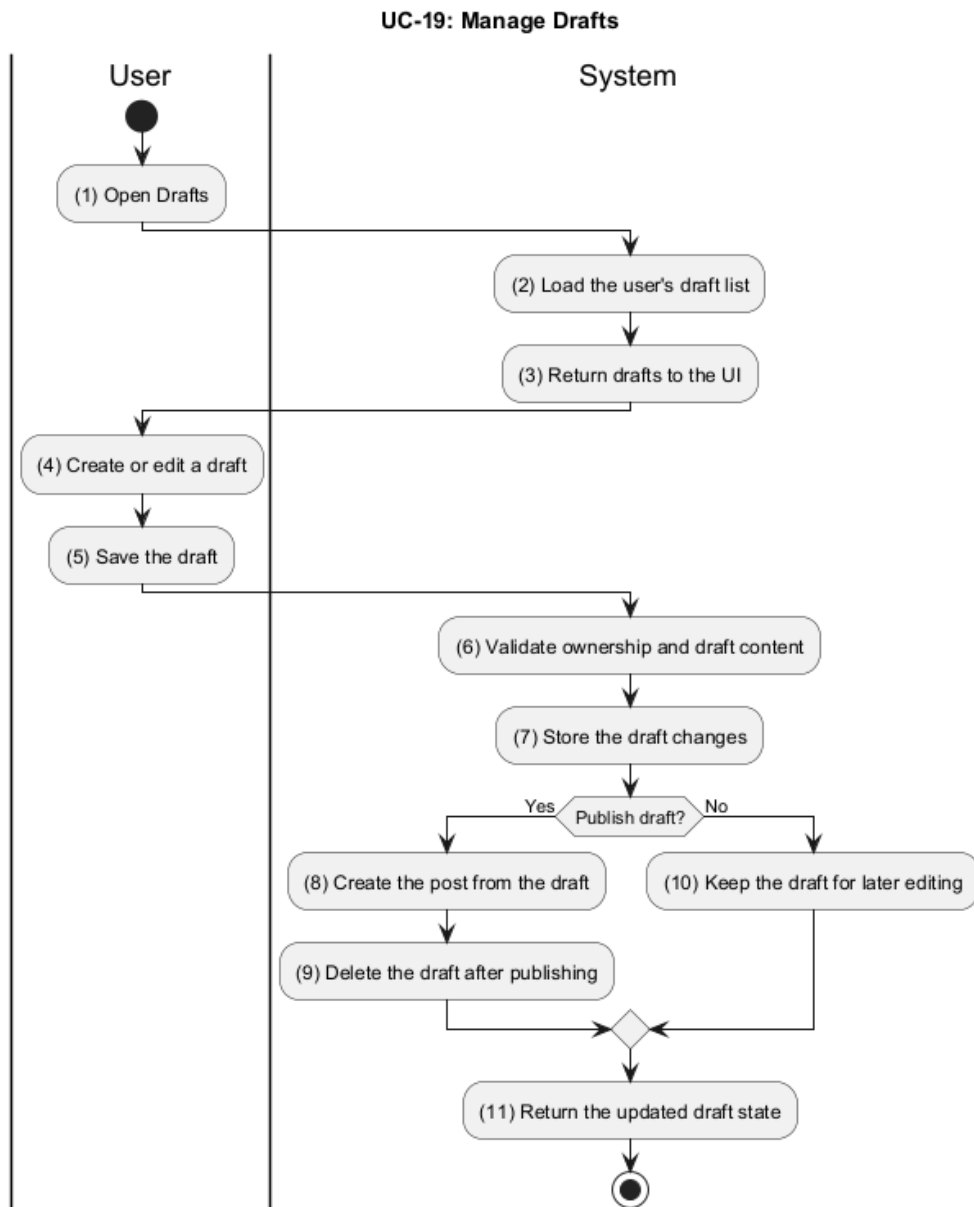


Figure 3.20: Activity Diagram for UC-19

Activity	BR Code	Description
(2-3)	BR1	Load Drafts: - The system loads the authenticated user's draft list using <i>GetDraftsByAuthor()</i>. - The system returns drafts together with pagination data.

Table 3.39: Business Rules for UC-19 - Manage Drafts

Activity	BR Code	Description
(6–7)	BR2	Create or Update Draft: • The system validates draft ownership before saving changes. • The system stores the draft via <i>CreateDraft()</i> or <i>UpdateDraft()</i>. • The system updates text, images, videos, poll data, tags, and community selection on the draft record.
(8–10)	BR3	Publish or Delete Draft: • The system checks required fields before publishing a draft. • The system creates the post from the draft using <i>PublishDraft()</i>. • After successful publishing, the system deletes the draft record.
(2–10)	BR4	Message Rules: • Draft actions shall use MSG-31, MSG-16, MSG-32, and MSG-86 where applicable. • Draft validation or lookup failures shall return ERR-58, ERR-26, ERR-10, ERR-39, or ERR-46.

Table 3.39: Business Rules for UC-19 - Manage Drafts

3.1.20 UC-20 - Private Messaging

Name	Private Messaging
Description	This use case describes how users exchange instant one-to-one messages through the messaging system.
Actor	User
Trigger	When the user opens a conversation and sends a message.
Pre-condition	The user is logged in to the system. The conversation channel exists and is accessible.
Post-condition	The message is delivered to the conversation members. The channel read state is updated.

Table 3.40: Use Case Specification - Private Messaging

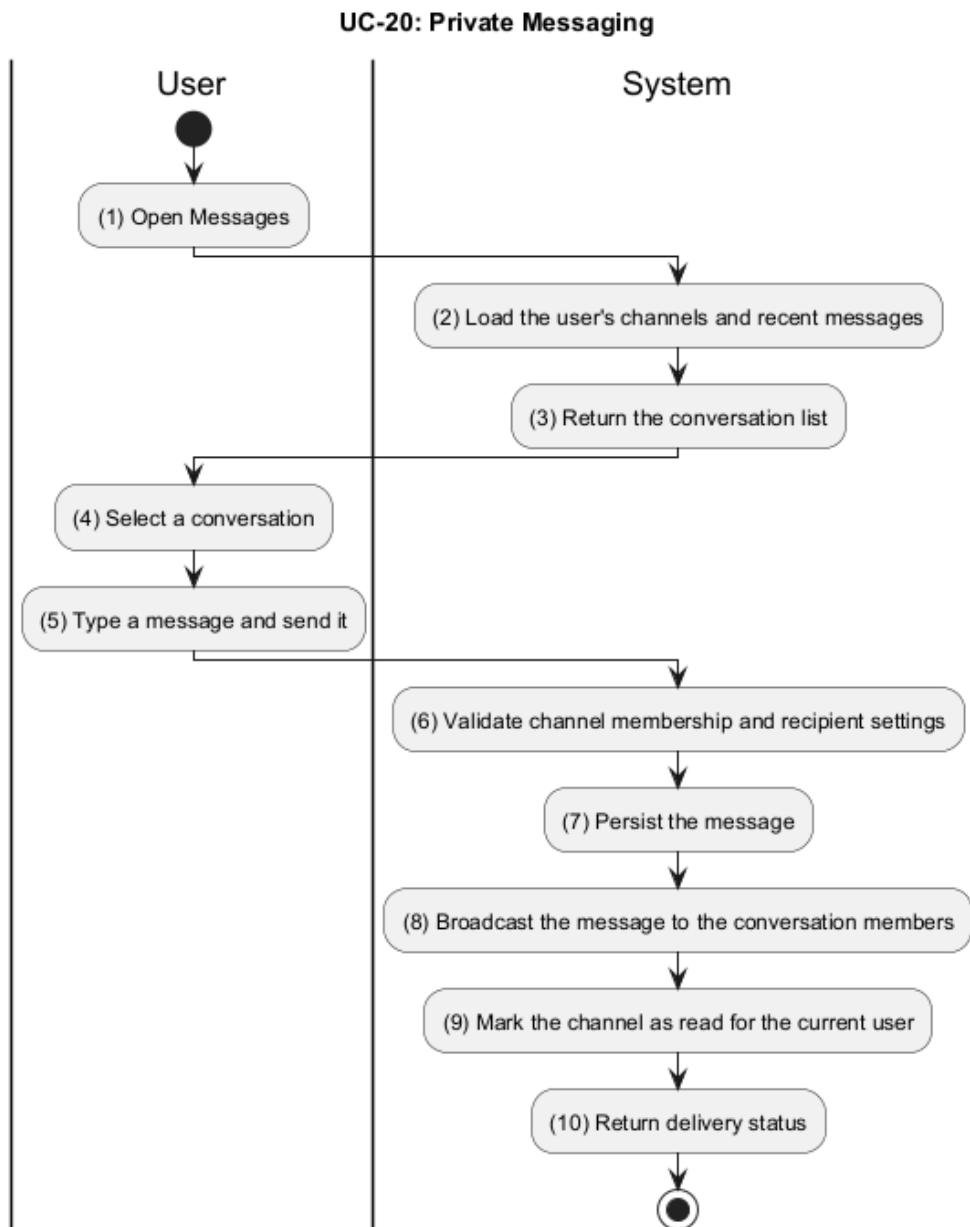


Figure 3.21: Activity Diagram for UC-20

Activity	BR Code	Description
(2-3)	BR1	Load Conversations: - The system loads the authenticated user's channels using <i>GetChannelsByUserID()</i>. - The system returns the conversation list and recent message state.

Table 3.41: Business Rules for UC-20 - Private Messaging

Activity	BR Code	Description
(4–10)	BR2	Send and Deliver Message: • The system validates that the sender is a channel member and that the recipients allow direct messages. • The system persists the message and broadcasts it to the other channel members. • The system marks the channel as read for the active user using <i>MarkChannelAsRead()</i>.
(2–10)	BR3	Message Rules: • Message-delivery failures shall use MSG-45, MSG-81, or MSG-82. • Channel and access failures shall return ERR-62, ERR-63, ERR-13, ERR-26, or ERR-10.

Table 3.41: Business Rules for UC-20 - Private Messaging

3.1.21 UC-21 - Manage Notifications

Name	Manage Notifications
Description	This use case describes how a user views and clears system notifications.
Actor	User
Trigger	When the user opens the notifications page or marks notifications as read.
Pre-condition	The user is logged in to the system. Notifications exist for the account.
Post-condition	The selected notifications are marked as read or deleted. The notification list is refreshed.

Table 3.42: Use Case Specification - Manage Notifications

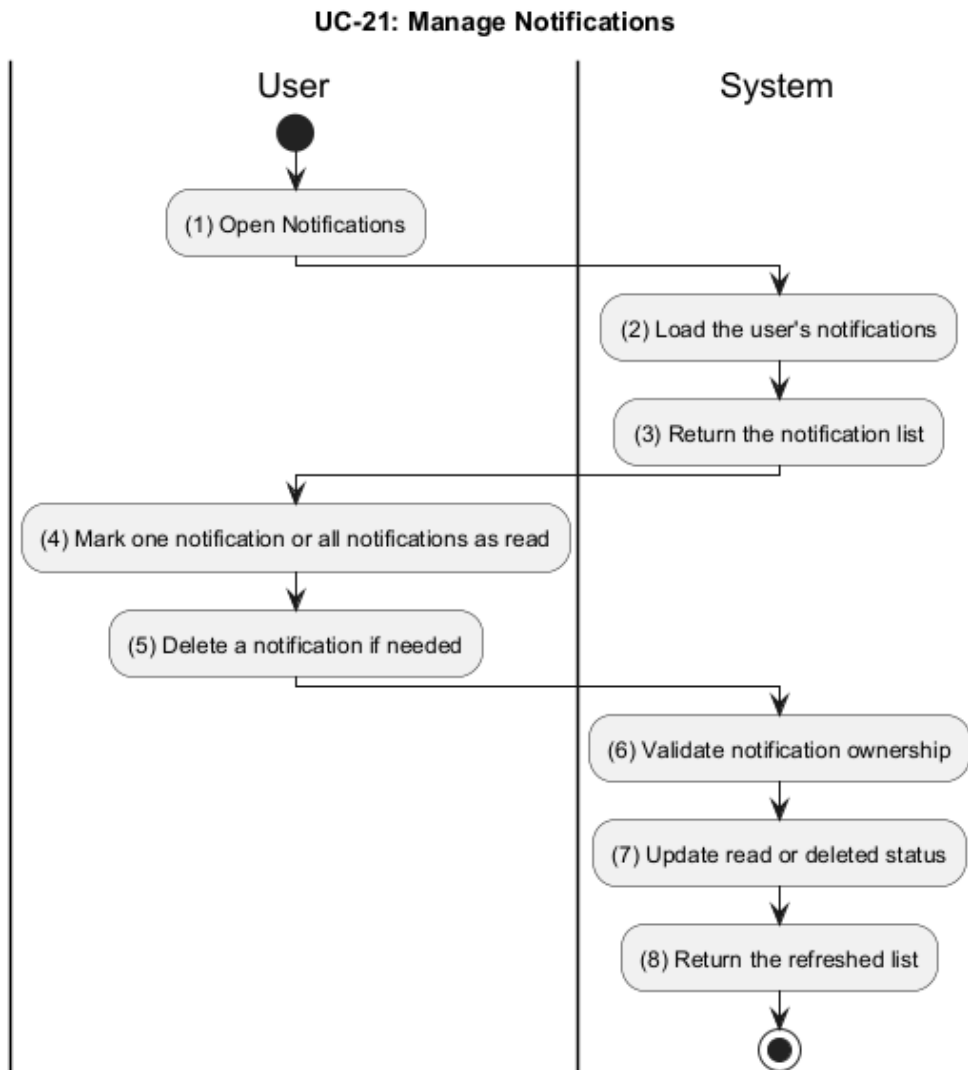


Figure 3.22: Activity Diagram for UC-21

Activity	BR Code	Description
(2–3)	BR1	Load Notifications: - The system loads notifications for recipientID using <i>GetNotifications()</i>. - The system returns the notification list with pagination data.

Table 3.43: Business Rules for UC-21 - Manage Notifications

Activity	BR Code	Description
(4-8)	BR2	Update Notification State: • The system validates notification ownership before changing any record. • If the user marks all notifications as read, the system updates them using <i>MarkAllAsRead()</i>. • If the user marks a single notification as read, the system validates notificationID, converts it to an object ID, and updates it using <i>MarkAsRead()</i>. • If the user deletes a notification, the system validates notificationID and removes it using <i>DeleteNotification()</i>. • The system returns the refreshed notification list after the update.
(2-8)	BR3	Message Rules: • Notification-related UI feedback shall use MSG-80, MSG-46, or MSG-47 when the notification action is an invitation. • Notification access or lookup failures shall return ERR-26, ERR-13, ERR-10, or ERR-2.

Table 3.43: Business Rules for UC-21 - Manage Notifications

3.1.22 UC-22 - Report Violation

Name	Report Violation
Description	This use case describes how a user reports inappropriate content or behavior for review.
Actor	User
Trigger	When the user submits a report form.
Pre-condition	The user is logged in to the system. The reported target is available.
Post-condition	The report is stored in the moderation queue. The report can be reviewed by moderators or admins.

Table 3.44: Use Case Specification - Report Violation

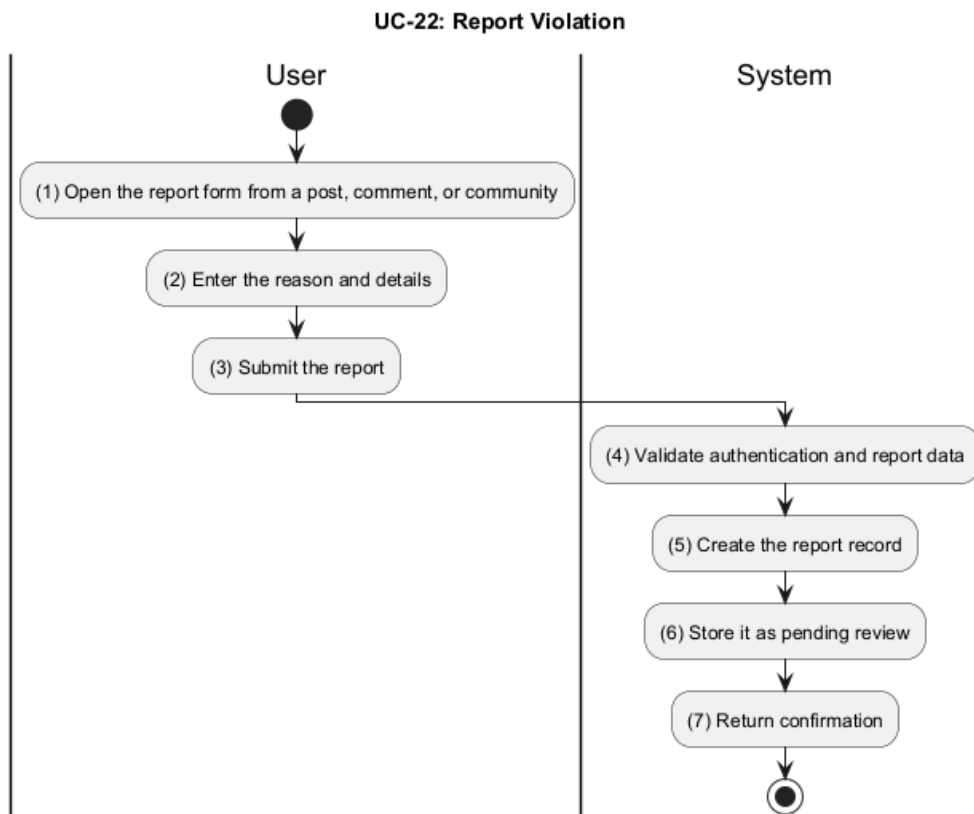


Figure 3.23: Activity Diagram for UC-22

Activity	BR Code	Description
(1–3)	BR1	Validate Report Submission: - The system checks that the user is authenticated. - The system validates the submitted reason and details in <i>ReportPost()</i>. - The system checks that the target post exists and that reporterID is not the post author.
(4–7)	BR2	Create Report Record: - The system creates a new report with Create() on the report repository. - The system stores the report as pending review. - The system returns a confirmation message after the report is saved.

Table 3.45: Business Rules for UC-22 - Report Violation

Activity	BR Code	Description
(1-7)	BR3	Message Rules: - Report submission shall use MSG-23 on success and MSG-24 on failure. - Report validation failures shall return ERR-57, ERR-53, ERR-59, ERR-28, ERR-26, or ERR-10.

Table 3.45: Business Rules for UC-22 - Report Violation

3.1.23 UC-23 - Create Community

Name	Create Community
Description	This use case describes how a user creates a new community and becomes its owner.
Actor	User
Trigger	When the user submits the create community form.
Pre-condition	The user is logged in to the system. The community name is available and valid.
Post-condition	The community is created successfully. The creator becomes the owner and first member.

Table 3.46: Use Case Specification - Create Community

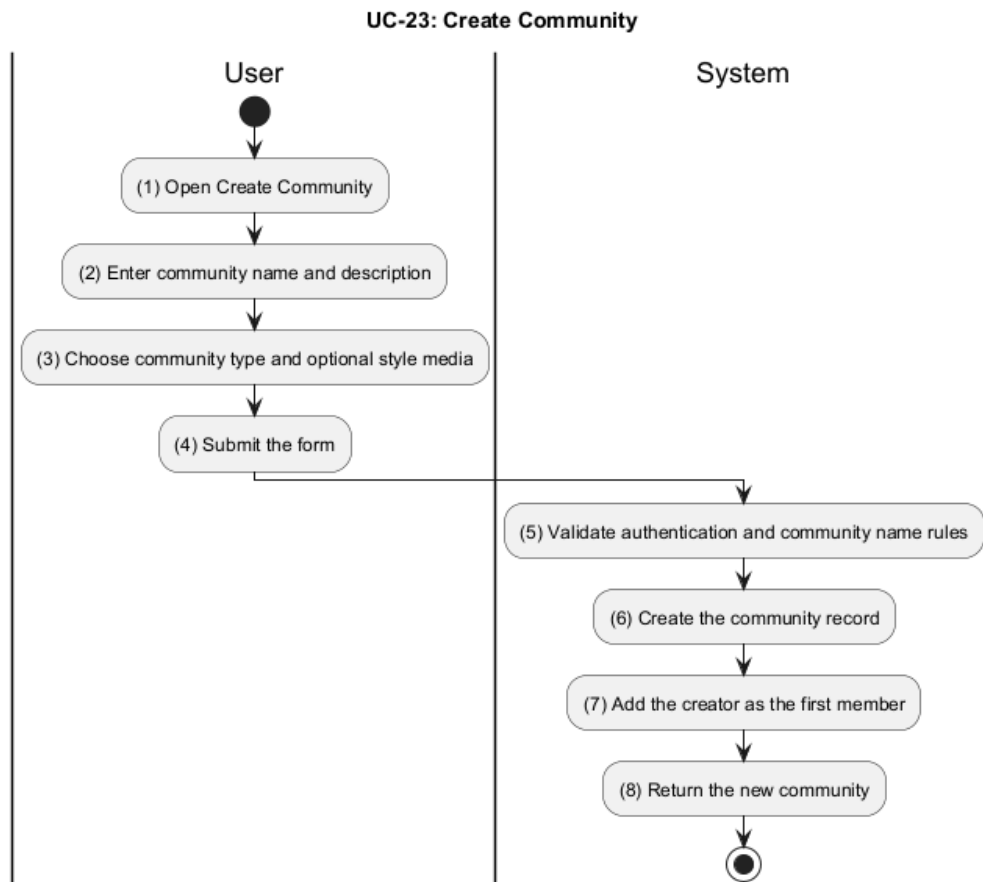


Figure 3.24: Activity Diagram for UC-23

Activity	BR Code	Description
(1–4)	BR1	Validate Community Input: - The system checks that the user is authenticated. - The system validates the submitted community fields, especially name in <i>CreateCommunity()</i>. - The system rejects duplicate community names with a community-name-exists error.
(5–8)	BR2	Create Community and Initial Membership: - The system creates the community record with <i>CreateCommunity()</i>. - The system adds the creator as the first member by creating a membership record. - The system returns the new community after persistence succeeds.

Table 3.47: Business Rules for UC-23 - Create Community

Activity	BR Code	Description
(1–8)	BR3	Message Rules: • Successful community creation shall use MSG-62. • Failed community creation shall use MSG-63. • Validation failures shall return ERR-40, ERR-26, ERR-10, or ERR-14.

Table 3.47: Business Rules for UC-23 - Create Community

3.1.24 UC-24 - Approve / Reject Post

Name	Approve / Reject Post
Description	This use case describes how a moderator reviews pending or edited posts and decides whether to publish them.
Actor	Moderator
Trigger	When the moderator opens the moderation queue and chooses approve or reject.
Pre-condition	The moderator is logged in to the system. The target post is pending review or edited.
Post-condition	The post is approved or rejected. The moderation result is stored and reflected in the community.

Table 3.48: Use Case Specification - Approve / Reject Post

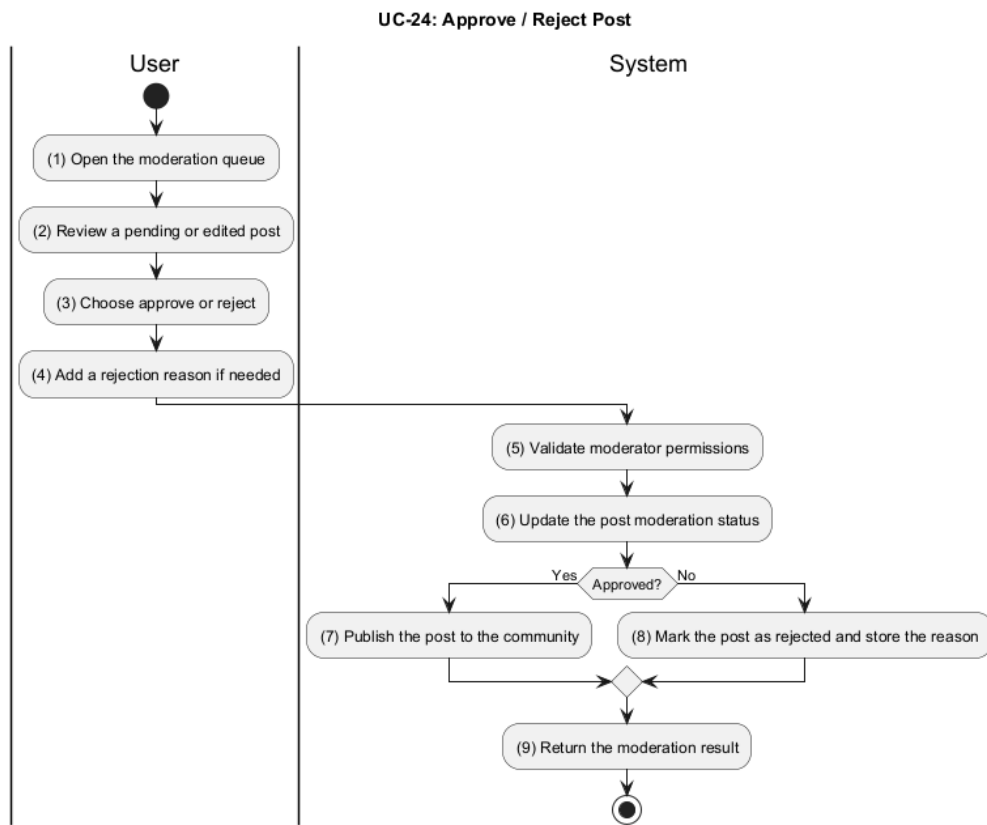


Figure 3.25: Activity Diagram for UC-24

Activity	BR Code	Description
(1–4)	BR1	Prepare Moderation Decision: - The system loads the moderation queue using <i>GetPendingPosts()</i> and <i>GetEditedPosts()</i>. - The system validates that the requester has moderator access. - The system checks that the selected post belongs to the current community and is still pending moderation.

Table 3.49: Business Rules for UC-24 - Approve / Reject Post

Activity	BR Code	Description
(5–8)	BR2	Apply Moderation Result: • The system updates the post using <i>ModeratePost()</i>. • If the moderator approves the post, the system sets the moderation status to approved and publishes the post to the community. • If the moderator rejects the post, the system sets the moderation status to rejected and stores the rejection reason. • The system returns the moderation result to the caller.
(1–8)	BR3	Message Rules: • Moderator approval shall use MSG-66; approval failures shall use MSG-67. • Rejection or removal flows may reuse MSG-87 and MSG-27 when the post is removed from the queue. • Moderation failures shall return ERR-53, ERR-13, ERR-39, or ERR-26.

Table 3.49: Business Rules for UC-24 - Approve / Reject Post

3.1.25 UC-25 - Ban / Mute User

Name	Ban / Mute User
Description	This use case describes how a moderator restricts a user's participation in a community.
Actor	Moderator
Trigger	When the moderator chooses to ban or mute a user.
Pre-condition	The moderator is logged in to the system. The target user belongs to the community.
Post-condition	The ban or mute restriction is stored. The user is prevented from performing the restricted actions.

Table 3.50: Use Case Specification - Ban / Mute User

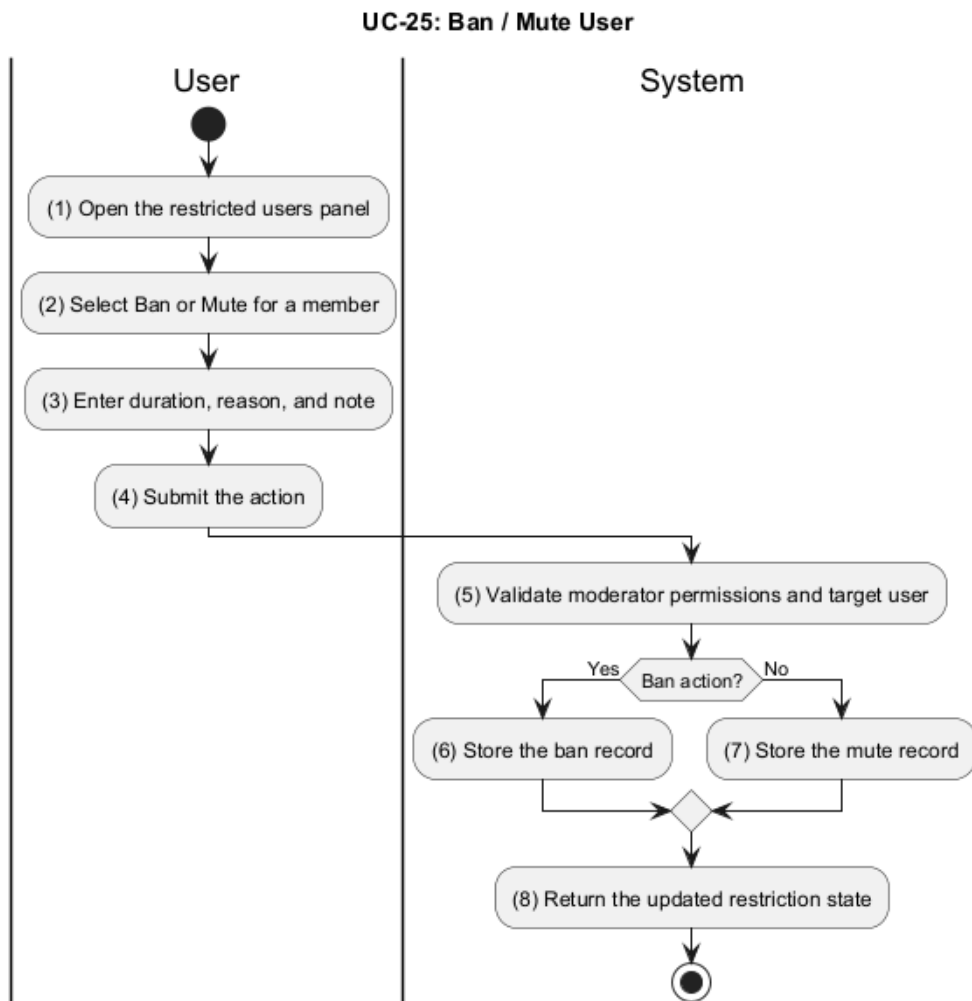


Figure 3.26: Activity Diagram for UC-25

Activity	BR Code	Description
(1–4)	BR1	Validate Restriction Request: • The system checks that the requester has moderator access. • The system validates the target userID, communityID, Type, Reason, and LengthDays. • The system accepts only the banned or muted restriction type.

Table 3.51: Business Rules for UC-25 - Ban / Mute User

Activity	BR Code	Description
(5-7)	BR2	Store Ban or Mute Record: - The system calculates the expiration timestamp from LengthDays. - The system creates a community ban record through <i>BanUser()</i>. - The system stores either a ban or mute entry based on Type and returns the updated restriction state.
(1-7)	BR3	Message Rules: - Restriction actions shall use MSG-68, MSG-69, MSG-70, MSG-71, MSG-72, or MSG-73 as applicable. - Validation or authorization failures shall return ERR-28, ERR-39, ERR-47, ERR-13, or ERR-26.

Table 3.51: Business Rules for UC-25 - Ban / Mute User

3.1.26 UC-26 - Configure Community

Name	Configure Community
Description	This use case describes how a community owner updates rules, privacy, and appearance settings.
Actor	Owner
Trigger	When the owner opens the community settings page and saves changes.
Pre-condition	The owner is logged in to the system. The owner has permission to manage the community.
Post-condition	The community configuration is updated. The new settings are applied to the community.

Table 3.52: Use Case Specification - Configure Community

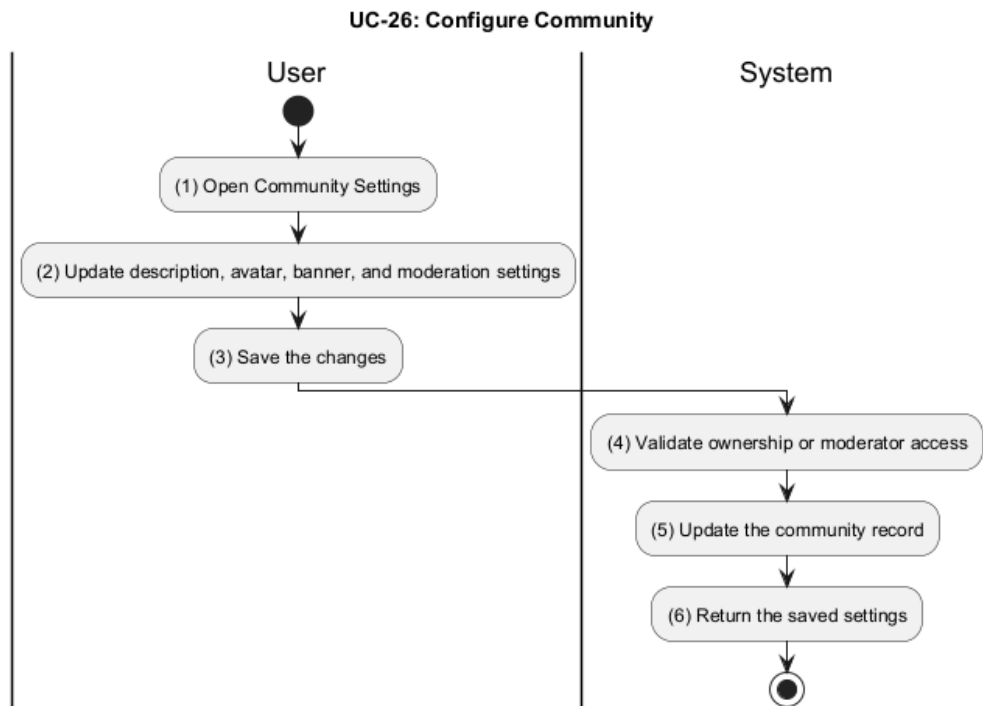


Figure 3.27: Activity Diagram for UC-26

Activity	BR Code	Description
(1–3)	BR1	Validate Community Access: - The system checks that the requester has moderator access for the community. - The system validates the incoming community settings before saving them.
(4–6)	BR2	Persist Community Settings: - The system updates the community record using <i>UpdateCommunity()</i>. - The system saves the description, avatar, banner, and moderation settings. - The system returns the saved community settings.

Table 3.53: Business Rules for UC-26 - Configure Community

Activity	BR Code	Description
(1–6)	BR3	Message Rules: • Successful community-configuration changes shall use MSG-74. • Failed updates shall use MSG-75. • Access or validation failures shall return ERR-39, ERR-13, ERR-26, or ERR-14.

Table 3.53: Business Rules for UC-26 - Configure Community

3.1.27 UC-27 - Assign Roles

Name	Assign Roles
Description	This use case describes how a community owner appoints moderators or other roles for members.
Actor	Owner
Trigger	When the owner invites or activates a role for another member.
Pre-condition	The owner is logged in to the system. The target member belongs to the community.
Post-condition	The role assignment is stored successfully. The target member gains the new permissions.

Table 3.54: Use Case Specification - Assign Roles

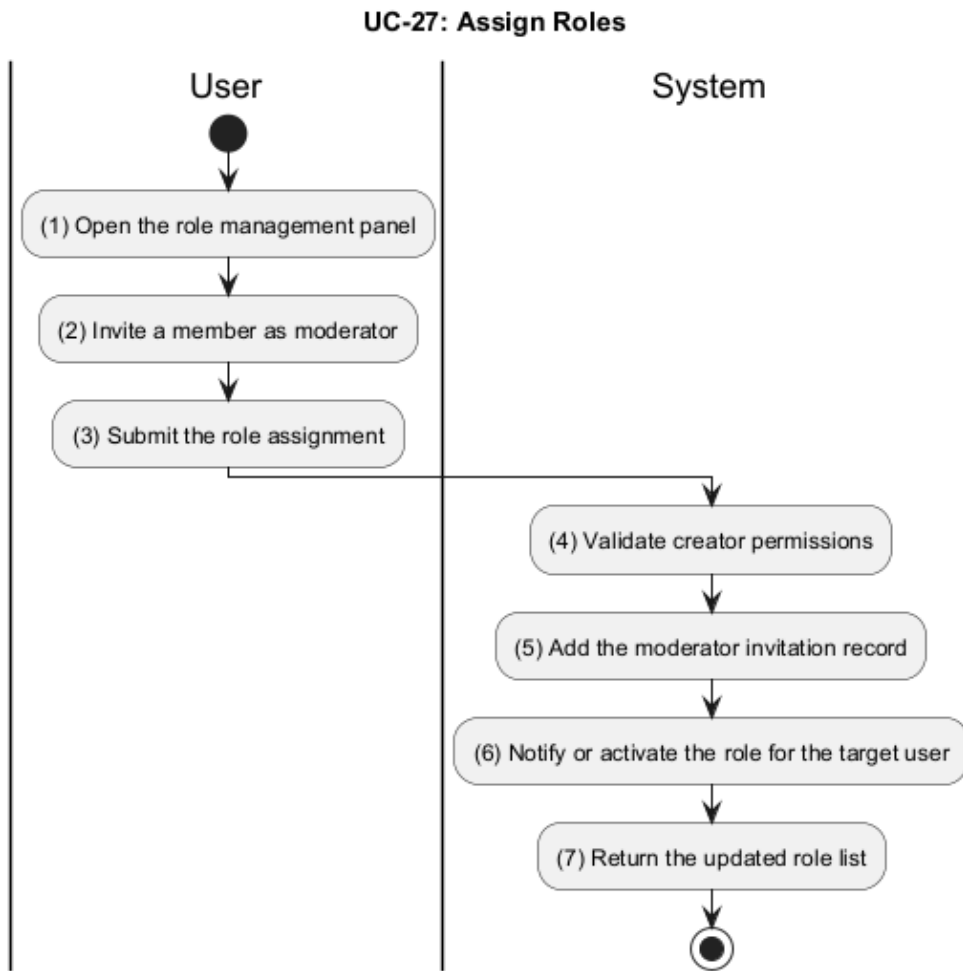


Figure 3.28: Activity Diagram for UC-27

Activity	BR Code	Description
(1–3)	BR1	Validate Role Assignment: - The system checks that the requester is the community creator. - The system validates the moderator assignment payload, including CommunityID and AddedModerator.

Table 3.55: Business Rules for UC-27 - Assign Roles

Activity	BR Code	Description
(4-7)	BR2	Add and Activate Moderator Role: - The system creates the moderator invitation or assignment record using <i>AddModerator()</i>. - The system activates the moderator role for the target user using <i>ActivateModerator()</i> when the role becomes active. - The system returns the updated moderator list.
(1-7)	BR3	Message Rules: - Role-assignment feedback shall use MSG-60 on success and MSG-61 on failure. - Validation failures shall return ERR-28, ERR-39, ERR-42, ERR-46, ERR-43, ERR-44, or ERR-26.

Table 3.55: Business Rules for UC-27 - Assign Roles

3.1.28 UC-28 - Handle Global Reports

Name	Handle Global Reports
Description	This use case describes how a system admin reviews and acts on platform-wide violation reports.
Actor	System Admin
Trigger	When the admin opens the global reports dashboard.
Pre-condition	The admin is logged in to the system. There are reports available for review.
Post-condition	The selected reports are reviewed or deleted. The report queue is updated.

Table 3.56: Use Case Specification - Handle Global Reports

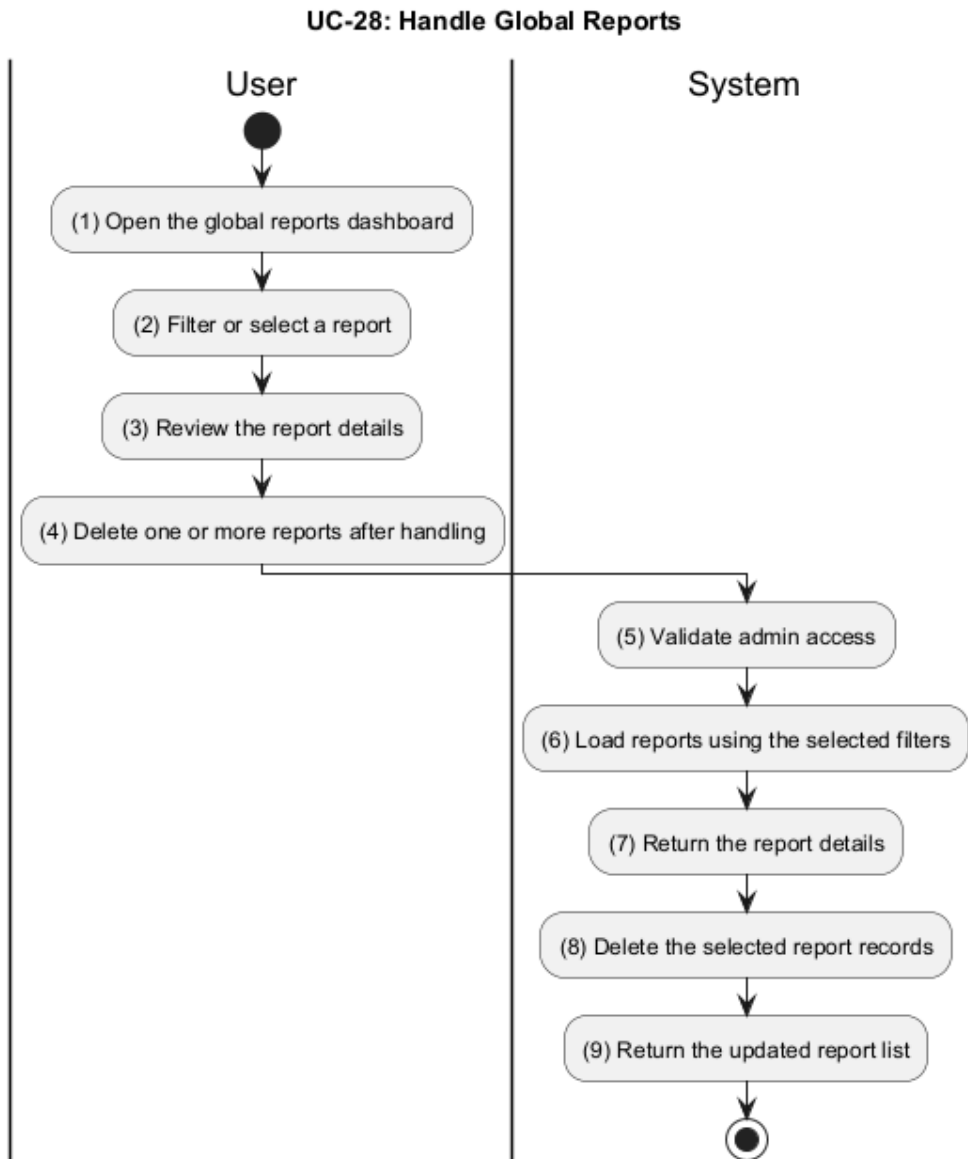


Figure 3.29: Activity Diagram for UC-28

Activity	BR Code	Description
(1–3)	BR1	Load and Inspect Reports: - The system checks that the requester is an admin. - The system loads filtered reports using <i>GetReportsFilter()</i>. - The system allows the admin to inspect a single report using <i>GetReportByID()</i>.

Table 3.57: Business Rules for UC-28 - Handle Global Reports

Activity	BR Code	Description
(4-9)	BR2	Delete Handled Reports: • The system deletes a single report using <i>DeleteReport()</i> or multiple reports using <i>DeleteReports()</i>. • The system validates that the admin is allowed to delete the selected report records. • The system rejects an empty batch request. • The system returns the updated report list after deletion.
(1-9)	BR3	Message Rules: • Report-administration UI feedback shall use MSG-76, MSG-18, MSG-43, or MSG-44. • Administrative failures shall return ERR-12, ERR-64, ERR-26, ERR-14, or ERR-13.

Table 3.57: Business Rules for UC-28 - Handle Global Reports

3.1.29 UC-29 - Manage Platform

Name	Manage Platform
Description	This use case describes how a system admin manages users and communities globally.
Actor	System Admin
Trigger	When the admin opens the platform management area and selects a target entity.
Pre-condition	The admin is logged in to the system. The admin has global management privileges.
Post-condition	The selected user or community is banned or unbanned. The platform state is updated accordingly.

Table 3.58: Use Case Specification - Manage Platform

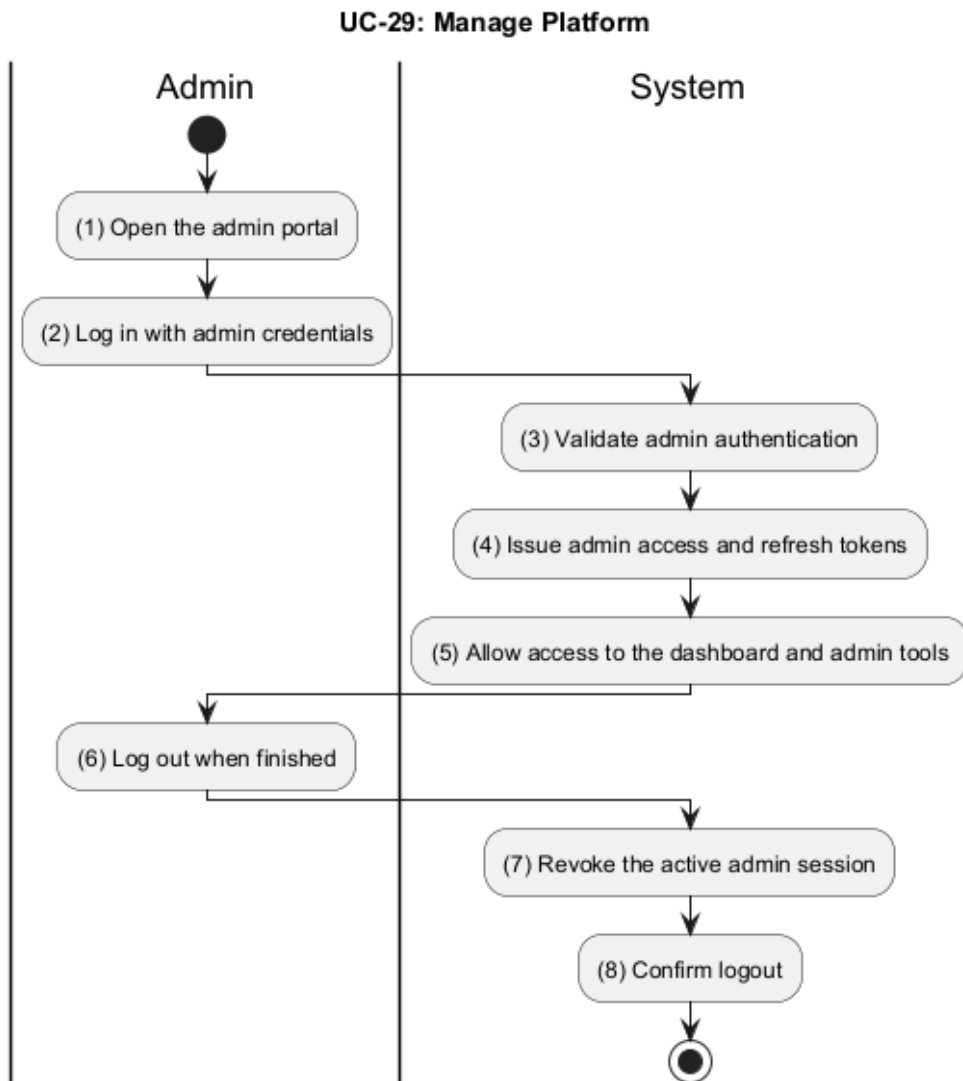


Figure 3.30: Activity Diagram for UC-29

Activity	BR Code	Description
(3)	BR1	Authenticate Admin Access: • The system checks the admin credentials and requires the user to have the admin role. • The system rejects banned or deleted admin accounts. • The system verifies the password for local accounts and issues access and refresh tokens using <i>AdminLogin()</i>.

Table 3.59: Business Rules for UC-29 - Manage Platform

Activity	BR Code	Description
(6–8)	BR2	End the Admin Session: - The system revokes the active admin session using <i>AdminLogout()</i>. - The system invalidates the stored access and refresh tokens when token revocation is available. - The system confirms logout after session cleanup.
(1–8)	BR3	Message Rules: - Platform-management UI feedback shall use MSG-77, MSG-78, MSG-33, MSG-34, MSG-35, MSG-36, MSG-37, MSG-38, MSG-39, MSG-40, MSG-41, or MSG-42. - Administrative failures shall return ERR-12, ERR-1, ERR-28, ERR-39, ERR-26, or ERR-2.

Table 3.59: Business Rules for UC-29 - Manage Platform

3.1.30 UC-30 - View Analytics

Name	View Analytics
Description	This use case describes how a system admin reviews platform usage and performance metrics.
Actor	System Admin
Trigger	When the admin opens the analytics dashboard.
Pre-condition	The admin is logged in to the system. Analytics data is available for the selected period.
Post-condition	The analytics dashboard is displayed with current metrics. The admin can review user, content, and moderation statistics.

Table 3.60: Use Case Specification - View Analytics

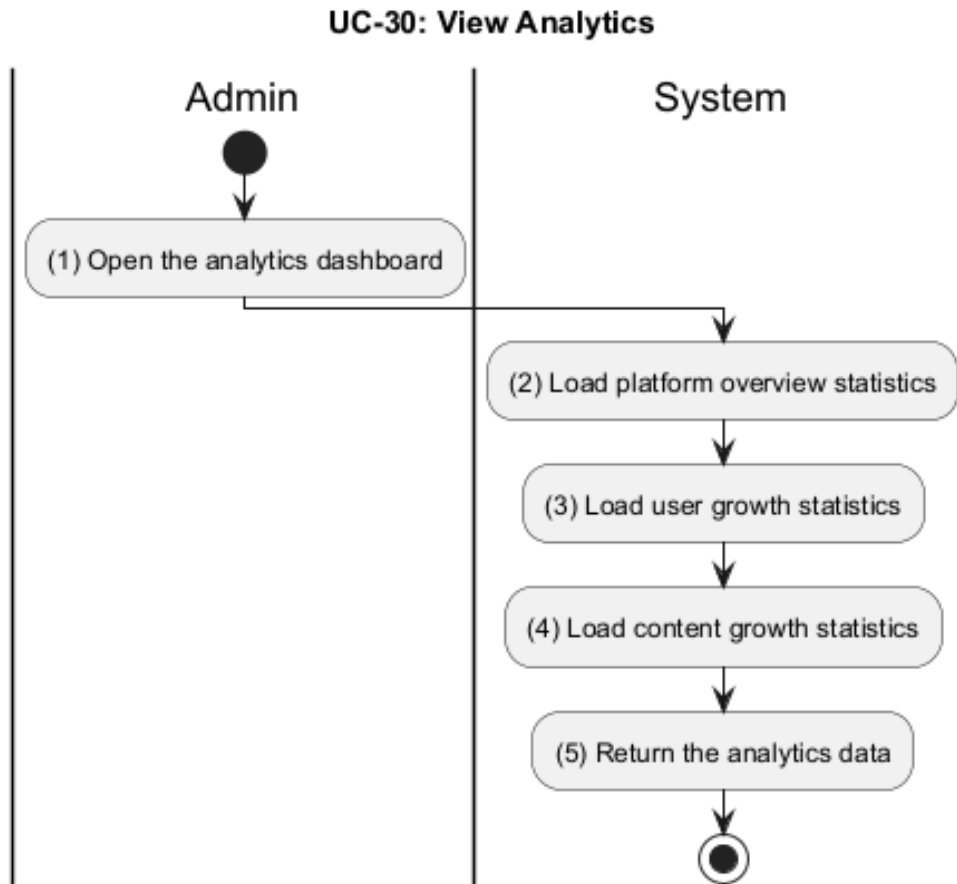


Figure 3.31: Activity Diagram for UC-30

Activity	BR Code	Description
(1–5)	BR1	Load Platform Analytics: - The system checks that the requester is an admin. - The system loads platform overview data using <i>GetPlatformOverview()</i>. - The system aggregates user, content, and moderation statistics from the backend services.
(2–5)	BR2	Return Analytics Breakdowns: - The system returns user growth and content growth data using <i>GetUserStats()</i> and <i>GetContentStats()</i>. - The system includes the current overview snapshot in the analytics response.

Table 3.61: Business Rules for UC-30 - View Analytics

3.2.2 Data Dictionary

Authentication and User Profile Models

Data Element	Description	Composition / Data Type	Length	Values
ID	Unique identifier of a user.	objectId	24	System-generated, unique.
Username	Display name used to identify a user account.	string	–	Must be unique within the platform.
Email	Email address used for authentication and contact.	string	–	Must be unique and valid.
Reputation	User reputation score.	integer	–	System-maintained counter.
Password	Hashed password for a local account.	string	–	Stored only for local providers.
Provider	Authentication source for a user account.	enum	–	local, google
ProviderID	External account identifier returned by the provider.	string	–	Used for third-party login linkage.
Role	Platform-wide role of the user.	enum	–	user, admin
RoleContent	Role-specific data attached to the user account.	object	–	Contains user or admin sub-document data.
Settings	User preference configuration.	object	–	Contains appearance, notification, privacy, and content settings.
IsVerified	Indicates whether the account has been verified.	boolean	–	true, false
IsBanned	Indicates whether the account is currently banned.	boolean	–	true, false
BanUntil	Timestamp when a ban expires, if any.	timestamp	–	Null means permanent ban.
BanReason	Reason recorded for a ban action.	string	–	Optional free text.
CreatedAt	Timestamp when the record was created.	timestamp	–	System-generated.
DeletedAt	Timestamp for soft-deleted records.	timestamp	–	Null when not deleted.

Table 3.62: User

Data Element	Description	Composition / Data Type	Length	Values
AsUser	User-specific profile data.	object	–	See UserRoleContent table.
AsAdmin	Admin-specific profile data.	object	–	See AdminRoleContent table.

Table 3.63: RoleContent

Data Element	Description	Composition / Data Type	Length	Values
Avatar	Profile image.	object	–	See Image table.
Cover	Cover image.	object	–	See Image table.
Bio	Short biography.	string	–	User-generated text.
Gender	Declared gender.	enum	–	male, female, prefer_not_to_say
DateOfBirth	User date of birth.	timestamp	–	Optional date value.
Location	Province or city.	enum	–	One of the provinces defined in model/location.go.
Interests	List of user interests.	array[string]	–	Values defined in model/interest.go.
SocialLinks	Collection of public social profiles.	object	–	Website, Facebook, YouTube, GitHub links.
Stats	User activity statistics.	object	–	Post, comment, and vote counters plus timestamps.

Table 3.64: UserRoleContent

Data Element	Description	Composition / Data Type	Length	Values
Website	Personal website URL.	string	–	Optional URL.
Facebook	Facebook profile URL.	string	–	Optional URL.
YouTube	YouTube channel URL.	string	–	Optional URL.
GitHub	GitHub profile URL.	string	–	Optional URL.

Table 3.65: SocialLinks

Data Element	Description	Composition / Data Type	Length	Values
PostCount	Number of posts created.	integer	–	System-maintained counter.
CommentCount	Number of comments created.	integer	–	System-maintained counter.
TotalUpvotes	Total upvotes received.	integer	–	System-maintained counter.
JoinedAt	Timestamp when the user joined.	timestamp	–	System-generated.
LastActiveAt	Timestamp of the last activity.	timestamp	–	System-generated.

Table 3.66: ActivityStats

Data Element	Description	Composition / Data Type	Length	Values
Permissions	Admin capability list.	array[string]	–	Permission names granted to the admin.

Table 3.67: AdminRoleContent

Data Element	Description	Composition / Data Type	Length	Values
Appearance	Visual preferences.	object	–	See AppearanceSettings table.
Notifications	Notification preferences.	object	–	See NotificationSettings table.
Privacy	Privacy preferences.	object	–	See PrivacySettings table.
Content	Content filtering preferences.	object	–	See ContentSettings table.

Table 3.68: UserSettings

Data Element	Description	Composition / Data Type	Length	Values
Theme	UI theme preference.	enum	–	light, dark, auto
FontSize	UI font size preference.	enum	–	small, medium, large

Table 3.69: AppearanceSettings

Data Element	Description	Composition / Data Type	Length	Values
InAppEnabled	In-app notification toggle.	boolean	–	true, false
EmailEnabled	Email notification toggle.	boolean	–	true, false
NotifyOn-Comment	Notify on comments.	boolean	–	true, false
NotifyOn-Mention	Notify on mentions.	boolean	–	true, false
NotifyOnUpvote	Notify on upvotes.	boolean	–	true, false
NotifyOnMessage	Notify on private messages.	boolean	–	true, false

Table 3.70: NotificationSettings

Data Element	Description	Composition / Data Type	Length	Values
ShowProfile	Controls whether a profile is public.	boolean	–	true, false
ShowEmail	Controls whether email is visible on the profile.	boolean	–	true, false
ShowPostHistory	Controls whether post history is visible.	boolean	–	true, false
AllowDirect-Messages	Controls whether direct messages are allowed.	boolean	–	true, false
AllowMentions	Controls whether mentions are allowed.	boolean	–	true, false

Table 3.71: PrivacySettings

Data Element	Description	Composition / Data Type	Length	Values
AllowNSFW	Controls whether NSFW content is allowed.	boolean	–	true, false

Table 3.72: ContentSettings

Data Element	Description	Composition / Data Type	Length	Values
EmailVerification-ID	Unique identifier of an email verification record.	objectId	24	System-generated, unique.

Table 3.73: Verification Records

EmailVerification- .Email	Email being verified.	string	–	Must be unique and valid.
EmailVerification- .OTP	One-time password used for verification.	string	6	Six-digit numeric code.
EmailVerification- .OTPExpiresAt	Expiration timestamp for the OTP.	timestamp	–	System-generated.
EmailVerification- .IsVerified	Indicates whether the OTP has already been verified.	boolean	–	true, false
EmailVerification- .Nonce	Anti-replay token.	string	–	System-generated random value.
EmailVerification- .CreatedAt	Timestamp when the record was created.	timestamp	–	System-generated.

Table 3.73: Verification Records

Data Element	Description	Composition / Data Type	Length	Values
PasswordReset- .ID	Unique identifier of a password reset record.	objectId	24	System-generated, unique.
PasswordReset- .Email	Email being reset.	string	–	Must be unique and valid.
PasswordReset- .OTP	One-time password used for reset.	string	6	Six-digit numeric code.
PasswordReset- .OTPExpiresAt	Expiration timestamp for the OTP.	timestamp	–	System-generated.
PasswordReset- .IsVerified	Indicates whether the OTP has already been verified.	boolean	–	true, false
PasswordReset- .Nonce	Anti-replay token.	string	–	System-generated random value.
PasswordReset- .CreatedAt	Timestamp when the record was created.	timestamp	–	System-generated.

Table 3.74: PasswordReset

Data Element	Description	Composition / Data Type	Length	Values
URL	Public URL of a stored image.	string	–	Cloudinary or equivalent media URL.

Table 3.75: Image

PublicID	Provider-specific image identifier.	string	–	Used for delete/update operations.
UploadedAt	Timestamp when the image was uploaded.	timestamp	–	System-generated.

Table 3.75: Image

Data Element	Description	Composition / Data Type	Length	Values
URL	Public URL of a stored video.	string	–	Cloudinary or equivalent media URL.
PublicID	Provider-specific video identifier.	string	–	Used for delete/update operations.
UploadedAt	Timestamp when the video was uploaded.	timestamp	–	System-generated.

Table 3.76: Video

Community and Membership Models

Data Element	Description	Composition / Data Type	Length	Values
ID	Unique identifier of a community.	objectId	24	System-generated, unique.
Name	Public name of a community.	string	–	Must be unique.
Description	Text describing a community.	string	–	Optional free text.
Avatar	Community avatar image URL.	string	–	Optional media URL.
Banner	Banner image for a community.	string	–	Optional media URL.
Setting	Community configuration settings.	object	–	Contains privacy and moderation flags.
Rules	Published rules of a community.	array[object]	–	Title and description entries.
Moderators	List of assigned community moderators.	array[object]	–	Moderator sub-documents.
MemberCount	Current number of community members.	integer	–	System-maintained counter.
PostCount	Current number of community posts.	integer	–	System-maintained counter.

Table 3.77: Community

CreateAt	Timestamp when the community was created.	timestamp	–	System-generated.
CreateByID	Identifier of the creator.	objectId	24	Must correspond to an existing user id.
CreateByName	Username of the creator.	string	–	Copied from the creator account.
CreateByAvatar	Avatar URL of the creator.	string	–	Copied from the creator profile.
Is18Plus	Marks the community as adult content.	boolean	–	true, false
IsDeleted	Indicates whether the community is soft deleted.	boolean	–	true, false
IsBanned	Indicates whether the community is banned.	boolean	–	true, false
BanReason	Reason recorded for a community ban.	string	–	Optional free text.

Table 3.77: Community

Data Element	Description	Composition / Data Type	Length	Values
IsPrivate	Controls whether the community is visible only to members.	boolean	–	true, false
PostRequire-Approval	Controls whether new posts require moderator approval.	boolean	–	true, false
JoinRequire-Approval	Controls whether join requests require approval.	boolean	–	true, false
MaxPostLength	Maximum allowed post length in characters.	integer	–	Positive integer.

Table 3.78: CommunitySetting

Data Element	Description	Composition / Data Type	Length	Values
Title	Short title for a community rule.	string	–	User-defined label.
Description	Detailed explanation of a community rule.	string	–	User-defined text.

Table 3.79: CommunityRule

Data Element	Description	Composition / Data Type	Length	Values
UserID	Identifier of the moderator account.	objectId	24	Must correspond to an existing user id.
Username	Moderator username.	string	–	Copied from the user account.
Avatar	Moderator avatar.	object	–	See Image table.
IsActive	Indicates whether the moderator role is active.	boolean	–	true, false
AssignedAt	Timestamp when the moderator role was assigned.	timestamp	–	System-generated.

Table 3.80: Moderator

Data Element	Description	Composition / Data Type	Length	Values
ID	Unique identifier of a membership record.	objectId	24	System-generated, unique.
UserID	Identifier of the member user.	objectId	24	Must correspond to an existing user id.
CommunityID	Identifier of the target community.	objectId	24	Must correspond to an existing community id.
Status	Membership review status.	enum	–	pending, approved, rejected
CreatedAt	Timestamp when the membership was created.	timestamp	–	System-generated.
User	Cached user summary for a membership entry.	object	–	Contains id, username, and avatar.

Table 3.81: Membership

Data Element	Description	Composition / Data Type	Length	Values
ID	Identifier of the membership user summary.	objectId	24	Must correspond to an existing user id.
Username	Username of the membership user summary.	string	–	Copied from the user account.

Table 3.82: UserInfo

Avatar	Avatar of the membership user summary.	object	–	See Image table.
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Table 3.82: UserInfo

Data Element	Description	Composition / Data Type	Length	Values
CommunityID	Identifier of the banned community.	objectId	24	Must correspond to an existing community id.
UserID	Identifier of the banned user.	objectId	24	Must correspond to an existing user id.
Type	Ban category applied by the community.	enum	–	banned, muted
Reason	Reason recorded for the ban or mute.	string	–	Optional free text.
BannedAt	Timestamp when the restriction was applied.	timestamp	–	System-generated.
BannedBy	Identifier of the moderator who applied the restriction.	objectId	24	Must correspond to an existing user id.
ExpiresAt	Timestamp when the restriction expires.	timestamp	–	Null for permanent restrictions.

Table 3.83: CommunityBan

Content and Moderation Models

Data Element	Description	Composition / Data Type	Length	Values
ID	Unique identifier of a post.	objectId	24	System-generated, unique.
AuthorID	Identifier of the post author.	objectId	24	Must correspond to an existing user id.
CommunityID	Identifier of the community that owns the post.	objectId	24	Must correspond to an existing community id.
Type	Type of post content.	enum	–	text, poll, video, image
Title	Title of a post.	string	–	Text entered by the author.
Content	Main post payload.	object	–	Contains text, images, videos, or poll data.

Table 3.84: Post

VotesCount	Vote summary for a post or comment.	object	–	Up and down counts.
CommentsCount	Number of comments on a post.	integer	–	System-maintained counter.
HotScore	Ranking score used for best sorting.	float	–	Calculated from votes and time decay.
CreatedAt	Timestamp when the post was created.	timestamp	–	System-generated.
UpdatedAt	Timestamp of the last content update.	timestamp	–	Null until updated.
IsDeleted	Indicates whether the record is soft deleted.	boolean	–	true, false
IsHidden	Indicates whether the post is hidden by its author.	boolean	–	true, false
IsEdited	Indicates whether the post has been edited.	boolean	–	true, false
Tags	Tags attached to a post.	array[string]	–	Free-form keywords.
IsDraft	Indicates whether the post is saved as a draft.	boolean	–	true, false
IsBan	Indicates whether a post is banned.	boolean	–	true, false
BanReason	Reason recorded for a post ban.	string	–	Optional free text.
ModerationStatus	Current AI or moderator moderation state.	enum	–	pending, approved, rejected, skipped
ModerationResult	Moderation analysis details.	object	–	Contains violation flag, confidence, categories, and reason.
ModeratedAt	Timestamp when moderation happened.	timestamp	–	Null until moderated.

Table 3.84: Post

Data Element	Description	Composition / Data Type	Length	Values
Text	Text body of a post.	string	–	Optional for non-text posts.
Images	Attached images in a post.	array[object]	–	Array of image metadata.
Videos	Attached videos in a post.	array[object]	–	Array of video metadata.

Table 3.85: PostContent

Poll	Poll payload attached to a post.	object	–	Question, options, votes, expiry, and multi-select flag.
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Table 3.85: PostContent

Data Element	Description	Composition / Data Type	Length	Values
Question	Question presented to poll voters.	string	–	Poll text.
Options	Available poll options.	array[object]	–	Each option has id, text, and votes.
TotalVotes	Total number of poll votes.	integer	–	System-maintained counter.
ExpiresAt	Poll expiry timestamp.	timestamp	–	Null when the poll never expires.
AllowMultiple	Controls whether voters may choose more than one option.	boolean	–	true, false

Table 3.86: Poll

Data Element	Description	Composition / Data Type	Length	Values
ID	Identifier of a poll option.	string	–	Unique within the poll.
Text	Display text of a poll option.	string	–	Option label.
Votes	Vote count for a poll option.	integer	–	System-maintained counter.

Table 3.87: PollOption

Data Element	Description	Composition / Data Type	Length	Values
Up	Number of upvotes.	integer	–	System-maintained counter.
Down	Number of downvotes.	integer	–	System-maintained counter.

Table 3.88: VotesCount

Data Element	Description	Composition / Data Type	Length	Values
IsViolation	Indicates whether the content violates policy.	boolean	–	true, false
Confidence	Confidence score returned by moderation.	float	–	0.0 to 1.0
Categories	Categories detected by moderation.	array[string]	–	Issue labels such as hate speech or violence.
Reason	Human-readable moderation explanation.	string	–	Localized explanation text.
CheckedText	Indicates whether text was analyzed.	boolean	–	true, false
CheckedMedia	Indicates whether media was analyzed.	boolean	–	true, false

Table 3.89: ModerationResult

Data Element	Description	Composition / Data Type	Length	Values
ID	Unique identifier of a comment.	objectId	24	System-generated, unique.
Author	Author details attached to a comment.	object	–	Contains id, username, and avatar.
PostID	Identifier of the parent post.	objectId	24	Must correspond to an existing post id.
ParentID	Identifier of the parent comment, if any.	objectId	24	Null for top-level comments.
Content	Comment body text.	string	–	User-generated text.
VotesCount	Vote summary for a comment.	object	–	Up and down counts.
CreatedAt	Timestamp when the comment was created.	timestamp	–	System-generated.
DeletedAt	Timestamp for soft-deleted comments.	timestamp	–	Null when not deleted.
IsDeleted	Indicates whether a comment is soft deleted.	boolean	–	true, false

Table 3.90: Comment

Data Element	Description	Composition / Data Type	Length	Values
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Table 3.91: CommentAuthor

ID	Identifier of the comment author.	objectId	24	Must correspond to an existing user id.
Username	Username of the comment author.	string	–	Copied from the user account.
Avatar	Avatar for the comment author.	object	–	Image metadata.

Table 3.91: CommentAuthor

Data Element	Description	Composition / Data Type	Length	Values
ID	Unique identifier of a draft.	objectId	24	System-generated, unique.
AuthorID	Identifier of the draft author.	objectId	24	Must correspond to an existing user id.
CommunityID	Identifier of the draft community, if any.	string	–	Optional community id string.
Type	Post type stored in a draft.	enum	–	text, poll, video, image
Title	Draft title.	string	–	Optional until publish time.
Content	Draft content payload.	object	–	Same shape as post content.
Tags	Tags saved with the draft.	array[string]	–	Optional free-form keywords.
CreatedAt	Timestamp when the draft was created.	timestamp	–	System-generated.
UpdatedAt	Timestamp when the draft was last updated.	timestamp	–	System-generated.

Table 3.92: Draft

Data Element	Description	Composition / Data Type	Length	Values
URL	Public URL of a stored image.	string	–	Cloudinary or equivalent media URL.
PublicID	Provider-specific image identifier.	string	–	Used for delete/update operations.
UploadedAt	Timestamp when the image was uploaded.	timestamp	–	System-generated.

Table 3.93: Image

Data Element	Description	Composition / Data Type	Length	Values
URL	Public URL of a stored video.	string	–	Cloudinary or equivalent media URL.
PublicID	Provider-specific video identifier.	string	–	Used for delete/update operations.
UploadedAt	Timestamp when the video was uploaded.	timestamp	–	System-generated.

Table 3.94: Video

Engagement and Activity Models

Data Element	Description	Composition / Data Type	Length	Values
ID	Unique identifier of a saved-post record.	objectId	24	System-generated, unique.
UserID	Identifier of the user who saved the post.	objectId	24	Must correspond to an existing user id.
PostID	Identifier of the saved post.	objectId	24	Must correspond to an existing post id.
SavedAt	Timestamp when a post was saved.	timestamp	–	System-generated.

Table 3.95: SavedPost

Data Element	Description	Composition / Data Type	Length	Values
ID	Unique identifier of a liked-post record.	objectId	24	System-generated, unique.
UserID	Identifier of the user who liked the post.	objectId	24	Must correspond to an existing user id.
PostID	Identifier of the liked post.	objectId	24	Must correspond to an existing post id.
LikedAt	Timestamp when a post was liked.	timestamp	–	System-generated.

Table 3.96: LikedPost

Data Element	Description	Composition / Data Type	Length	Values
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Table 3.97: PostHistory

ID	Unique identifier of a post history record.	objectId	24	System-generated, unique.
UserID	Identifier of the user who viewed the post.	objectId	24	Must correspond to an existing user id.
PostID	Identifier of the viewed post.	objectId	24	Must correspond to an existing post id.
ViewedAt	Timestamp when a post was viewed.	timestamp	–	System-generated.

Table 3.97: PostHistory

Communication Models

Data Element	Description	Composition / Data Type	Length	Values
ID	Unique identifier of a notification.	objectId	24	System-generated, unique.
RecipientID	Identifier of the notification recipient.	objectId	24	Must correspond to an existing user id.
ActorID	Identifier of the user who triggered the notification.	objectId	24	Optional when the system is the actor.
Type	Type of notification.	enum	–	comment, like, follow, mention, new_message, system
Message	Notification text shown to the recipient.	string	–	System-generated or templated text.
Link	URL or route associated with the notification.	string	–	Navigation target.
IsRead	Indicates whether a notification has been read.	boolean	–	true, false
Metadata	Additional notification payload.	Dictionary	–	Arbitrary key-value data.
CreatedAt	Timestamp when the notification was created.	timestamp	–	System-generated.

Table 3.98: Notification

Data Element	Description	Composition / Data Type	Length	Values
ID	Unique identifier of a private message channel.	objectId	24	System-generated, unique.
Members	Channel participants.	array[object]	–	Member summaries.

Table 3.99: Channel

Settings	Per-user channel settings.	array[object]	–	Nickname and notification preferences.
Status	Channel lifecycle status.	enum	–	active, block
CreatedAt	Timestamp when the channel was created.	timestamp	–	System-generated.
UpdatedAt	Timestamp when the channel was last updated.	timestamp	–	System-generated.

Table 3.99: Channel

Data Element	Description	Composition / Data Type	Length	Values
UserID	Identifier of the user owning the channel settings.	objectId	24	Must correspond to a channel member.
Nickname	Custom nickname for the other user in the channel.	string	–	Optional display label.
Notification	Controls notification delivery for the channel.	boolean	–	true, false
TypingIndicator	Controls typing indicator visibility.	boolean	–	true, false
IsDeleted	Indicates whether the channel setting is hidden.	boolean	–	true, false

Table 3.100: ChannelSetting

Data Element	Description	Composition / Data Type	Length	Values
UserID	Identifier of the channel member.	objectId	24	Must correspond to an existing user id.
Username	Username of the channel member.	string	–	Copied from the user account.
Avatar	Avatar URL of the channel member.	string	–	Optional media URL.

Table 3.101: ChannelMember

Data Element	Description	Composition / Data Type	Length	Values
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Table 3.102: Message

ID	Unique identifier of a private message.	objectId	24	System-generated, unique.
ChannelID	Identifier of the private message channel.	objectId	24	Must correspond to an existing channel id.
SenderID	Identifier of the message sender.	objectId	24	Null for system messages.
SenderUsername	Cached sender username.	string	–	Copied from the sender account.
Type	Message category.	enum	–	user, system
Content	Message body text.	string	–	User-generated or system-generated text.
IsSend	Indicates whether the message was delivered successfully.	boolean	–	true, false
IsDeleted	Indicates whether the message is soft deleted.	boolean	–	true, false
DeletedAt	Timestamp when the message was deleted.	timestamp	–	Null when not deleted.

Table 3.102: Message

Reporting and Administration Models

Data Element	Description	Composition / Data Type	Length	Values
ID	Unique identifier of a report.	objectId	24	System-generated, unique.
ReporterID	Identifier of the user who submitted the report.	objectId	24	Must correspond to an existing user id.
TargetID	Identifier of the reported user, post, or comment.	objectId	24	Must correspond to an existing target id.
TargetType	Type of reported target.	enum	–	user, post, comment
Reason	Short reason for a report or moderation action.	string	–	Free text provided by the user or moderator.
Description	Longer report details or explanation.	string	–	Optional free text.
IsDeleted	Indicates whether the report is soft deleted.	boolean	–	true, false
DeletedAt	Timestamp for deleted reports.	timestamp	–	Null when not deleted.
CreatedAt	Timestamp when the report was created.	timestamp	–	System-generated.

Table 3.103: Report

Enum and Value

Data Element	Description	Composition / Data Type	Length	Values
AuthProvider	User authentication source.	enum	–	local, google
Role	Platform-wide user role.	enum	–	user, admin
Gender	Declared gender.	enum	–	male, female, prefer_not_to_say
VNProvince	User location value.	enum	–	One of the provinces defined in model/location.go.
Interest	User interest label.	enum	–	Values defined in model/interest.go.
PostType	Post content type.	enum	–	text, poll, video, image
Moderation-Status	Moderation state.	enum	–	pending, approved, rejected, skipped
Membership-Status	Membership review state.	enum	–	pending, approved, rejected
NotificationType	Notification type.	enum	–	comment, like, follow, mention, new_message, system
ChannelStatus	Channel lifecycle status.	enum	–	active, block
MessageType	Message category.	enum	–	user, system
ReportTarget-Type	Reported target type.	enum	–	user, post, comment
VoteType	Vote direction type.	enum	–	up, down
VoteTargetType	Vote target type.	enum	–	post, comment

Table 3.104: Enum and Value Classes

3.3 Data Integrity, Retention, and Disposal

- DI-1:** The system shall retain user-deleted content (such as posts and comments) in a deactivated state for 30 days. After this period, the content will be permanently purged from the system. This allows users a grace period to recover accidentally deleted items.
- DI-2:** When a user requests to delete their account, the account shall be immediately deactivated and scheduled for permanent deletion. All personally identifiable information and associated content will be permanently deleted or anonymized after 90 days, unless required to be kept for legal reasons.
- DI-3:** Moderation data, including content reports submitted by users and action logs taken by moderators, shall be retained for one year after the report is closed to monitor for patterns of abuse.
- DI-4:** Core content such as active posts and comments will be retained indefinitely as long as the author's account and the community they belong to exist.

4 Other Nonfunctional Requirements

4.1 Performance Requirements

- PR-1:** For normal operations, page navigation and standard read requests such as browsing posts, opening a post, loading a profile, and retrieving notifications should return the first visible result within 3 seconds under normal network conditions.
- PR-2:** Search requests across posts, users, and communities should return results within 5 seconds for typical keyword queries, with pagination applied to large result sets.
- PR-3:** Write operations such as creating posts, comments, votes, reports, and direct messages should return a confirmation response within 3 seconds after server validation completes, excluding third-party network delays.
- PR-4:** Time-consuming external tasks, including OTP email delivery and media upload processing, shall be handled asynchronously whenever possible so the user is not blocked longer than necessary.
- PR-5:** The system shall support concurrent use by end users, moderators, and administrators without data corruption in shared resources such as post counts, membership records, reports, notifications, and message channels.

4.2 Security Requirements

- SR-1:** All authenticated operations shall require valid access credentials, and protected endpoints shall reject requests with missing, expired, invalid, or revoked tokens.
- SR-2:** The system shall enforce role-based authorization for user, moderator, community owner, and administrator functions, including moderation queues, role assignment, report handling, and platform management.
- SR-3:** Sensitive account recovery flows shall use time-limited OTP values and verification tokens, and the system shall reject expired or mismatched OTP submissions.
- SR-4:** The system shall verify resource ownership before allowing updates or deletion of posts, comments, drafts, notifications, reports, channels, and user-specific settings.
- SR-5:** User passwords shall never be stored or transmitted in plain text, and password reset operations shall require successful OTP verification before a new password is accepted.
- SR-6:** Uploaded content such as avatars, covers, post images, and post videos shall be validated before storage to reduce the risk of invalid or malicious files entering the system.
- SR-7:** Communication between browser clients, frontend applications, backend services, and third-party identity providers shall be protected using secure transport protocols in production deployment.

4.3 Software Quality Attributes

- QA-1: Availability** – LKForum should remain available for continuous daily use, including account access, browsing, community interaction, moderation, and private messaging features.
- QA-2: Reliability** – The system shall preserve data consistency for community membership, moderation status, notifications, votes, reports, and message delivery even when operations are retried or partially fail.
- QA-3: Usability** – The user interface shall provide clear validation, confirmation, and error messages for common actions such as sign up, login, posting, reporting, moderation decisions, and password recovery.
- QA-4: Maintainability** – The software should remain modular across frontend, backend, and supporting services so that authentication, community management, moderation, messaging, and analytics can evolve independently.
- QA-5: Scalability** – The architecture should support growth in users, communities, posts, comments, notifications, and direct messages by relying on pagination, soft deletion, and external media storage services.
- QA-6: Compatibility** – The web application shall operate correctly on current versions of Chrome, Firefox, Safari, and Edge as defined in the operating environment.

5 Other Requirements

- OR-1:** The system shall integrate with third-party services required by release 1.0, including Google OAuth for social login and external email delivery for OTP-based verification and password recovery.
- OR-2:** Media assets uploaded by users shall be stored through an external media service so that application servers are not responsible for long-term binary file storage.
- OR-3:** The system shall maintain audit-relevant records for moderation actions, reports, bans, mutes, and administrative changes in accordance with the retention requirements defined in Section 3.5.
- OR-4:** The application shall continue to support both regular end-user workflows and administrative workflows through separate interfaces while using the same platform data and authorization model.
- OR-5:** The specification, user-facing messages, and operational behavior should remain consistent across future releases so business rules, activity diagrams, and test cases can be traced back to stable identifiers.

6 Appendix

6.1 Messages

Message Code	Message Content	Button
MSG-1	Verification code sent to your email. Please check your inbox.	–
MSG-2	A new verification code has been sent to your email.	–
MSG-3	Login successful	–
MSG-4	Email verified successfully. You can now complete your registration.	–
MSG-5	Registration completed successfully. You are now logged in.	–
MSG-6	Password reset code sent to your email. Please check your inbox.	–
MSG-7	OTP verified successfully. You can now reset your password.	–
MSG-8	Password reset successfully. You can now login with your new password.	–
MSG-9	Mã OTP đã được gửi đến email của bạn!	–
MSG-10	Mã OTP mới đã được gửi đến email của bạn!	–
MSG-11	Xác thực email thành công!	–
MSG-12	Đăng ký thành công! Chào mừng bạn đến với LKForum.	–
MSG-13	Đổi mật khẩu thành công! Vui lòng đăng nhập lại.	–
MSG-14	Bạn có chắc chắn muốn xóa bài viết này? Hành động này không thể hoàn tác.	Xóa/Hủy
MSG-15	Bạn có chắc chắn muốn xóa bình luận này? Hành động này không thể hoàn tác.	Xóa/Hủy
MSG-16	Bạn có chắc chắn muốn xóa bản nháp này?	Xóa/Hủy
MSG-17	Bạn có chắc chắn muốn xóa người dùng này? Hành động này không thể hoàn tác.	Xóa/Hủy
MSG-18	Bạn có chắc chắn muốn xóa báo cáo này?	Xóa/Hủy
MSG-19	Bạn có chắc chắn muốn rời khỏi lk/<community-name>?	Rời/Hủy
MSG-20	Bạn có chắc chắn muốn bỏ các thay đổi? Các thay đổi chưa lưu sẽ bị mất.	Bỏ thay đổi/Tiếp tục chỉnh sửa

Message Code	Message Content	Button
MSG-21	Đã sao chép liên kết!	—
MSG-22	Không thể sao chép liên kết	—
MSG-23	Đã gửi báo cáo thành công	—
MSG-24	Không thể gửi báo cáo. Vui lòng thử lại.	—
MSG-25	Không thể gửi phiếu bầu. Vui lòng thử lại.	—
MSG-26	Không thể hủy phiếu bầu. Vui lòng thử lại.	—
MSG-27	Không thể xóa bài viết. Vui lòng thử lại.	—
MSG-28	Không thể xóa bình luận. Vui lòng thử lại.	—
MSG-29	Cập nhật bài viết thành công	—
MSG-30	Không thể cập nhật bài viết. Vui lòng thử lại.	—
MSG-31	Đã lưu bản nháp!	—
MSG-32	Không thể xóa bản nháp. Vui lòng thử lại.	—
MSG-33	Đã cấm người dùng	—
MSG-34	Không thể cấm người dùng	—
MSG-35	Đã gỡ cấm người dùng	—
MSG-36	Không thể gỡ cấm người dùng	—
MSG-37	Đã xóa người dùng	—
MSG-38	Không thể xóa người dùng	—
MSG-39	Đã cấm cộng đồng	—
MSG-40	Không thể cấm cộng đồng	—
MSG-41	Đã gỡ cấm cộng đồng	—
MSG-42	Không thể gỡ cấm cộng đồng	—
MSG-43	Đã xóa báo cáo	—
MSG-44	Không thể xóa báo cáo	—
MSG-45	Không thể gửi tin nhắn. Vui lòng thử lại.	—
MSG-46	Bạn đã chấp nhận làm moderator!	—
MSG-47	Không thể chấp nhận lời mời	—

Message Code	Message Content	Button
MSG-48	Đã cập nhật hồ sơ!	—
MSG-49	Không thể cập nhật hồ sơ.	—
MSG-50	Không thể đổi mật khẩu.	—
MSG-51	Đã lưu	—
MSG-52	Không thể lưu cài đặt.	—
MSG-53	Đã tải ảnh đại diện lên!	—
MSG-54	Không thể tải ảnh đại diện.	—
MSG-55	Đã xóa ảnh đại diện!	—
MSG-56	Không thể xóa ảnh đại diện.	—
MSG-57	Bạn có chắc muốn xóa ảnh đại diện?	Xóa/Hủy
MSG-58	Không thể tải cộng đồng	—
MSG-59	Đã tham gia cộng đồng <community-name>!	—
MSG-60	Đã thêm <username> làm quản trị viên!	—
MSG-61	Không thể thêm quản trị viên. Vui lòng kiểm tra tên người dùng và thử lại.	—
MSG-62	Đã tạo cộng đồng "lk/<community-name>" thành công!	—
MSG-63	Không thể tạo cộng đồng. Vui lòng thử lại.	—
MSG-64	Không thể đăng bình luận. Vui lòng thử lại.	—
MSG-65	Không thể gửi trả lời. Vui lòng thử lại.	—
MSG-66	Post approved successfully!	—
MSG-67	Failed to approve post. Please try again.	—
MSG-68	User banned successfully!	—
MSG-69	Đã tắt tiếng người dùng thành công!	—
MSG-70	Đã bỏ cấm người dùng thành công!	—
MSG-71	Không thể bỏ cấm người dùng. Vui lòng thử lại.	—
MSG-72	Đã bỏ tắt tiếng người dùng thành công!	—
MSG-73	Không thể bỏ tắt tiếng người dùng. Vui lòng thử lại.	—
MSG-74	Đã lưu cài đặt thành công!	—

Message Code	Message Content	Button
MSG-75	Không thể lưu cài đặt	—
MSG-76	Không thể tải danh sách báo cáo	—
MSG-77	Không thể tải danh sách người dùng	—
MSG-78	Không thể tải danh sách cộng đồng	—
MSG-79	Không thể tải thống kê	—
MSG-80	Không thể tải thông báo	—
MSG-81	Failed to create channel: <error>	—
MSG-82	Không thể mở trò chuyện. Vui lòng thử lại.	—
MSG-83	Không thể tải ảnh đại diện. Vui lòng thử lại.	—
MSG-84	Không thể tải bài viết. Vui lòng thử lại.	—
MSG-85	Không thể tải bài viết đã lưu. Vui lòng thử lại.	—
MSG-86	Failed to load draft. Please try again.	—
MSG-87	Đã xóa bài viết thành công	—

Table 6.1: Application Messages

6.2 Error Codes

Error ID	Code	Message
ERR-1	INVALID_CREDENTIALS	Email hoặc mật khẩu không đúng
ERR-2	INVALID_TOKEN	Token không hợp lệ hoặc đã hết hạn
ERR-3	INVALID_CLAIMS	Thông tin token không hợp lệ
ERR-4	INVALID_ISSUER	Nguồn phát hành token không hợp lệ
ERR-5	INVALID_AUDIENCE	Đối tượng token không hợp lệ
ERR-6	TOKEN_INVALIDATED	Token đã bị vô hiệu hóa
ERR-7	MISSING_AUTH_HEADER	Thiếu Authorization header
ERR-8	INVALID_AUTH_HEADER	Định dạng Authorization header không hợp lệ
ERR-9	MISSING_TOKEN	Thiếu token xác thực

Error ID	Code	Message
ERR-10	NOT_AUTHENTICATED	Chưa xác thực
ERR-11	INVALID_AUTH_- CONTEXT	Ngữ cảnh xác thực không hợp lệ
ERR-12	ADMIN_ACCESS_- REQUIRED	Cần quyền quản trị viên
ERR-13	FORBIDDEN	Bạn không có quyền thực hiện hành động này
ERR-14	BAD_REQUEST	Yêu cầu không hợp lệ
ERR-15	EMAIL_NOT_VERIFIED	Email chưa được xác thực
ERR-16	EMAIL_ALREADY_- VERIFIED	Email đã được xác thực
ERR-17	INVALID_OTP	Mã xác thực không đúng
ERR-18	OTP_EXPIRED	Mã xác thực đã hết hạn
ERR-19	LOGIN_METHOD_- MISMATCH	Tài khoản này đã đăng ký bằng Google. Vui lòng đăng nhập bằng Google hoặc liên hệ quản trị viên để hỗ trợ.
ERR-20	EMAIL_NOT_- REGISTERED	Email không tồn tại trong hệ thống
ERR-21	INVALID_USERNAME	Username phải từ 3-20 ký tự
ERR-22	PASSWORD_TOO_- SHORT	Mật khẩu phải có ít nhất 8 ký tự
ERR-23	PASSWORD_TOO_WEAK	Mật khẩu phải chứa ít nhất 1 chữ hoa, 1 chữ thường, 1 số và 1 ký tự đặc biệt (@\$!*%?&)
ERR-24	INTERNAL_ERROR	Lỗi hệ thống
ERR-25	NO_FIELDS_TO_- UPDATE	Không có trường nào để cập nhật
ERR-26	INVALID_ID	Định dạng ID không hợp lệ
ERR-27	PAGINATION_INVALID	Số trang hoặc kích thước trang không hợp lệ. Kích thước trang phải nhỏ hơn 500.
ERR-28	USER_NOT_FOUND	Không tìm thấy người dùng
ERR-29	USERNAME_EXISTS	Tên người dùng đã tồn tại
ERR-30	EMAIL_EXISTS	Email đã được sử dụng
ERR-31	USER_INACTIVE	Tài khoản người dùng đã bị vô hiệu hóa
ERR-32	INVALID_GENDER	Giá trị giới tính không hợp lệ
ERR-33	INVALID_DATE_- FORMAT	Định dạng ngày không hợp lệ, sử dụng YYYY-MM-DD
ERR-34	AGE_TOO_YOUNG	Phải từ 13 tuổi trở lên
ERR-35	INVALID_BIRTH_DATE	Ngày sinh không hợp lệ

Error ID	Code	Message
ERR-36	INVALID_PROVINCE	Tỉnh/thành phố không hợp lệ
ERR-37	TOO_MANY_INTERESTS	Tối đa 10 sở thích
ERR-38	INVALID_INTEREST	Sở thích không hợp lệ
ERR-39	COMMUNITY_NOT_FOUND	Không tìm thấy cộng đồng
ERR-40	COMMUNITY_NAME_EXISTS	Tên cộng đồng đã tồn tại
ERR-41	COMMUNITY_DELETED	Cộng đồng đã bị xóa
ERR-42	MODERATOR_ALREADY_EXISTS	Người dùng đã là moderator của cộng đồng
ERR-43	CANNOT_REMOVE_MODERATOR	Không thể xóa moderator này
ERR-44	CANNOT_REMOVE_CREATOR	Không thể xóa người tạo cộng đồng khỏi danh sách moderator
ERR-45	BANNED_COMMUNITY	Người dùng đã bị ban khỏi cộng đồng
ERR-46	USER_NOT_MEMBER	Bạn chưa tham gia cộng đồng này
ERR-47	MEMBERSHIP_NOT_FOUND	Không tìm thấy thành viên
ERR-48	ALREADY_MEMBER	Bạn đã là thành viên của cộng đồng này
ERR-49	MEMBERSHIP_CREATE_FAILED	Không thể tạo thành viên
ERR-50	MEMBERSHIP_DELETE_FAILED	Không thể xóa thành viên
ERR-51	INVALID_MEMBERSHIP_DATA	Dữ liệu thành viên không hợp lệ
ERR-52	MEMBERSHIP_ALREADY_PROCESSED	Yêu cầu tham gia đã được xử lý
ERR-53	POST_NOT_FOUND	Không tìm thấy bài viết
ERR-54	VOTE_NOT_FOUND	Không tìm thấy bình chọn
ERR-55	POLL_ALREADY_VOTED	Bạn đã bình chọn lựa chọn này rồi
ERR-56	POLL_CANNOT_EDIT	Không thể chỉnh sửa bình chọn sau khi đã có người bình chọn
ERR-57	ALREADY_REPORTED	Bạn đã báo cáo nội dung này rồi
ERR-58	DRAFT_NOT_FOUND	Không tìm thấy bản nháp
ERR-59	COMMENT_NOT_FOUND	Không tìm thấy bình luận
ERR-60	DEPTH_TOO_HIGH	Độ sâu phải từ 0 đến 2

Error ID	Code	Message
ERR-61	USER_MUTED	Người dùng đã bị cấm bình luận
ERR-62	CHANNEL_NOT_FOUND	Không tìm thấy kênh
ERR-63	NO_MESSAGE_FOUND	Không tìm thấy tin nhắn
ERR-64	REPORT_NOT_FOUND	Không tìm thấy report

Table 6.2: Application Error Codes